



A Comprehensive Bibliometric-Systematic Literature Review (B + SLR) of Microlearning Research: Mapping Global Trends, Applications, and Challenges in Education and Training

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Received: 23 May 2025 / Accepted: 6 February 2026 / Published online: 19 February 2026

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Abstract

Microlearning has emerged as a flexible and scalable pedagogical approach for knowledge acquisition and professional skill development in contemporary education. This study presents a bibliometric–systematic literature review (B + SLR) to examine global research trends in microlearning, with a focus on higher education, corporate training, and health education. Following PRISMA guidelines, 49 high-impact studies were selected from the Web of Science and Scopus databases using predefined inclusion and exclusion criteria. Bibliometric analyses were conducted using RStudio and Biblioshiny, incorporating co-word analysis, co-citation mapping, and thematic evolution techniques. The results indicate a steady growth in microlearning research over the past decade, with dominant themes including mobile learning, gamification, and artificial intelligence–enabled personalization. The United States leads in publication output and citation impact, while Iran and Malaysia also demonstrate notable contributions, reflecting increasing global engagement. However, adoption challenges persist, particularly in resource-constrained contexts, due to technological limitations and pedagogical readiness. The study highlights the need for improved digital infrastructure, educator capacity-building, and inclusive microlearning design to support scalable, equitable implementation.

Keywords Microlearning · B + SLR · Mobile learning · Digital education · Global trends · Flexible learning · Quality education

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Introduction

Microlearning: Concept and Definitions

In the rapidly evolving digital landscape, microlearning has emerged as a transformative educational paradigm, particularly within the realms of professional training and higher education (Leong et al., 2021). This approach segments extensive instructional content into small, digestible units delivered through concise, targeted formats, thereby enhancing learners' ability to comprehend, retain, and apply complex concepts effectively (Sankaranarayanan et al., 2023). Its prominence has intensified, driven by the proliferation of mobile technologies and the widespread adoption of online learning platforms. These factors collectively enable flexible, on-demand access to educational resources anytime and anywhere.

Despite numerous efforts to define microlearning (bite-sized or just-in-time learning), there is still no commonly accepted definition. Cronin and Durham (2024) define microlearning as an autonomous, demand-driven instructional approach that delivers concise, single-topic, interactive content to learners asynchronously, supported by multimodal delivery. Other definitions describe microlearning as an informal or formal learning process that occurs in bite-sized steps, supported by clear and manageable nuggets or pieces of content and activities (Horst et al., 2020). Briefly, microlearning stands for not only a self-learning method, but also a form of social learning offered in easily digestible formats with small units.

Importantly, microlearning involves designing self-contained, focused learning units that provide meaningful and coherent learning experiences within a limited scope, rather than merely fragmenting existing instructional content. Earlier studies have found that it encourages self-directed learning (Kohnke et al., 2024), fosters social connectedness (Torgerson, 2021), improves critical thinking skills (Fidan, 2023), and enhances motivation and learning performance (Kohnke, 2023). Moreover, microlearning offers various potential benefits for personalized and adaptive learning experiences (Alias & Razak, 2024). On the other hand, as with any instructional approach, there are several challenges, such as eye strain, distraction, poorly designed micro-content, insufficient interaction (Díaz Redondo et al., 2021), lack of contextual depth, and ambiguous instructional design (Lohman, 2024). Despite its growing popularity, sustaining student engagement remains difficult, and establishing consistent microlearning habits poses a significant challenge (Draxler et al., 2022). Hence, it is important to strategically and systematically integrate microlearning into the teaching and learning process.

Characteristics and Formats of Microlearning

Microlearning is not a new concept and has gained popularity with the advancements in mobile and internet technologies (Moore et al., 2024). Owing to its asynchronous nature, microlearning leverages electronic devices such as computers or mobile phones to facilitate learning. The main characteristics of microlearning include “time” (minimal time consumption and effort), “content” (short units and well-defined subject matter), “curriculum” (modular components and brief didactic elements), “format” (fragments, facets, episodes, lab assignments, etc.), “process” (iterative progression, networked patterns, and attention management strategies), “media” (printed, digital, television, etc.), and “learning model” (self-directed, concept-based, and connectivist approaches) (Hug, 2006, 2012). Cronin and Durham (2024) identify seven attributes commonly associated with microlearning: single focus (limited learning objectives and simple units), bite-sized (small chunks of information, units, or lessons), asynchronous (self-directed, user-driven learning), accessibility (available around the clock), flexibility (independent of time and location), interactivity (use of gamification, quizzes, etc.), and multimodal delivery (via electronic media or software platforms).

As for its format, microlearning encompasses a range of media from images and infographics to short videos, podcasts, flashcards, mobile apps, mini-games, audio, flip-books, mind maps, and animated GIFs (Kohnke, 2023; Wagener, 2021). In addition to traditional formats, cutting-edge technologies such as artificial intelligence (AI), augmented reality (AR), and virtual reality (VR) enhance microlearning by providing more interactive and immersive learning experiences, allowing learners to engage in realistic simulations applicable to educational and professional training contexts. Microlearning emphasizes repetition and conciseness to deliver focused and easily memorable information, maximizing impact quickly (Leong et al., 2021). This approach underscores the importance of just-in-time learning in improving information retention (Giurgiu, 2017; Shail, 2019).

Research by De Gagne et al. (2019a) indicates that microlearning reduces cognitive load and enhances long-term retention compared to traditional learning methods. The advancement of technology, such as smartphones and learning management systems (LMS), accelerates the adoption of microlearning across various sectors, including corporate training and higher education. These technologies enable learners to study according to their schedules while fostering more flexible and adaptive distance learning (Hug, 2010; Taylor & Hung, 2022).

Theoretical Foundations

Today's learners increasingly prefer brief, focused lessons that address a single concept and can be consumed efficiently. The adaptability and effectiveness of microlearning are particularly well-suited to Generation Z (Gen Z), individuals who have grown up in a digital environment and exhibit a strong preference for fast-paced, interactive, and easily accessible technology-enhanced learning (Harari et al., 2023; Seemiller & Grace, 2017; Weng & Seemiller, 2024). This generation values immediacy and convenience in accessing information, favoring short, on-demand content that minimizes waiting and maximizes engagement (Dziuban et al., 2018). Recent studies also highlight Gen Z's openness to change, enthusiasm for adopting emerging technologies such as generative AI, and preference for flexible, personalized learning experiences (Chan & Lee, 2023; Chardonens, 2025). Consistent with these characteristics, prior research underscores the importance of speed, accessibility, and user experience in sustaining learner engagement and satisfaction within digital environments (Rita et al., 2019). Slow-loading or poorly designed content not only hinders engagement but also parallels challenges in educational settings, where lengthy or unfocused instruction can significantly diminish learner motivation and participation (Nkomo et al., 2021; Shail, 2019; Stone, 2021).

Seemiller and Grace (2018) found that more than 70% of Gen Z individuals prioritize personalized learning methods, enabling quick access and alignment with their dynamic lifestyles, in line with heutagogical principles (Hernandez-de-Menendez et al., 2020; Tolstikova et al., 2021). Technologies such as social media, exemplified by Instagram, and streaming platforms such as TikTok have influenced people's consumption habits, favoring modular and audio-visual content with short durations. This context positions microlearning as an ideal choice for them (Conde-Caballero et al., 2024; Mayer, 2009). With its flexibility, personalization, and socialization, microlearning addresses these needs by providing adaptive learning experiences tailored to the digital characteristics of Gen Z.

Beyond its alignment with generational learning preferences, microlearning is grounded in robust cognitive theories that address the limitations of human memory (Stone, 2021). According to the "Forgetting Curve" introduced by Ebbinghaus (2013), individuals tend to forget most of the newly learned information within the first few days unless it is periodically reinforced. Microlearning leverages the "Spacing Effect," where information is presented in small chunks and repeated at specific intervals, which has proven effective in enhancing memory retention (Dempster, 1989; Hintzman et al., 1975).

Furthermore, microlearning alleviates cognitive load by breaking down complex concepts into smaller, more

digestible units (Fisher & Radvansky, 2019). This approach aligns directly with Sweller's Cognitive Load Theory (Sweller, 2024) and is also supported by the "Chunking Theory," which asserts that individuals process and retain information more efficiently when it is presented in discrete, well-organized segments (Cowan, 2001; Miller, 1956). These theoretical foundations reinforce the pedagogical value of microlearning, particularly in digital environments, by enhancing comprehension and long-term retention (Gobet, 2005). Emerging evidence also suggests that when micro-units are delivered repeatedly over time, they significantly improve knowledge consolidation and recall (Hug, 2006).

In addition to its cognitive underpinnings, microlearning has demonstrated considerable effectiveness in addressing educational disruptions during global crises. During the COVID-19 pandemic, digital learning methods such as microlearning became the primary solution for maintaining educational continuity when face-to-face learning was impossible (Salta et al., 2022). According to UNESCO (2020a), over 1.6 billion learners were affected by school closures, and many institutions shifted to digital solutions, including microlearning, to continue educational activities. This approach is relevant during crises and serves as a flexible and scalable learning solution in the long term (UNESCO, 2020b). More than 60% of educational institutions in developed countries have begun transitioning to microlearning to support adult education and lifelong learning (McKinsey, 2022).

Limitations of Existing Reviews

Previous research has shown that microlearning is rapidly emerging as a promising and adaptive solution in digital education, especially in the post-pandemic era, where technology adoption is increasingly widespread. Studies such as those by Leong et al. (2021) and Fialho et al. (2024) indicate a rising trend in microlearning adoption across various sectors, with particular emphasis on higher education and professional training. This body of research also underscores the significance of digital platforms, such as mobile devices and e-learning tools, in supporting flexible and accessible learning (Samala et al., 2023a, 2023b). Additionally, studies by De Gagne et al. (2019b) and Alias and Razak (2023) have examined the role of microlearning in healthcare professions and the pedagogical design of higher education, further reinforcing its relevance in both formal and informal learning contexts.

Despite these contributions, prior reviews exhibit distinct and isolated limitations that hinder a holistic understanding of microlearning research. Systematic and scoping reviews, for instance, often focus narrowly on specific contexts. De Gagne et al. (2019a) examined microlearning in healthcare education, finding positive effects on knowledge and

confidence (Kirkpatrick levels 1 and 2), yet almost no evidence on behavioral change (level 3) or patient outcomes (level 4). Similarly, Alias and Razak (2023) investigated pedagogical design in higher education but remained context-specific, limiting insight into broader adoption trends. Existing bibliometric studies, while offering a macro-level perspective, often provide surface-level analyses of publication metrics without integrating qualitative interpretations of pedagogical implications or theoretical developments. For example, Leong et al. (2021) mapped publication trends and online search interest to identify microlearning as an emerging topic, yet did not explore the underlying thematic evolution, and Fialho et al. (2024) highlighted geographic biases but remained focused on publication patterns rather than conceptual depth.

Additional limitations include a predominant focus on developed countries (Fialho et al., 2024) and limited exploration of challenges and contextual solutions in developing nations (Sankaranarayanan et al., 2023). Moreover, many studies concentrate on specific sectors, such as music or healthcare (De Gagne et al., 2019b; Mazlan et al., 2023), resulting in a lack of comprehensive insight into microlearning adoption and implementation across diverse industries and educational settings. Consequently, a synthesis that connects macro-level publication patterns with micro-level thematic interpretations is largely absent, highlighting the need for a holistic and methodologically integrated approach to understanding global microlearning research.

Rationale and Research Questions

To address these gaps, this study applies advanced science mapping using RStudio and the Biblioshiny package (Aria & Cuccurullo, 2017; Guleria & Kaur, 2021), implementing a hybrid approach that combines bibliometric analysis with systematic literature review (SLR), referred to as B + SLR (Samala et al., 2023a, 2023b). This analytical framework integrates findings from SLR, scoping reviews, and bibliometric studies with empirical evidence from primary research to provide a comprehensive analysis of trends and

developments in the field. Although the B + SLR approach remains underutilized in microlearning research, it offers novel insights with minimal bias. The analysis adheres to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2010; Page et al., 2021), ensuring transparency and systematic rigor in the selection, screening, and reporting of integrated study results. A detailed comparison of bibliometric analysis and SLR, including purpose, analytical focus, methods, and expected outcomes, is provided in Table 1 in the Methodology section.

The B + SLR conducted in this study examines global developments and emerging trends in microlearning. This approach involves analyzing publication patterns, identifying key contributors, and highlighting influential institutions shaping the field (Donthu et al., 2021). Additionally, citation analysis and productivity assessments were performed to uncover the central themes that support the flexibility, adaptability, and scalability of microlearning across various educational and professional domains. To address the core challenges and strategic considerations related to microlearning implementation, four primary research questions (RQs) were formulated:

- 1) RQ1: How have global trends in the adoption and implementation of microlearning evolved across sectors based on the B + SLR method?
- 2) RQ2: Who are the key contributors and institutions influencing the development of the microlearning literature?
- 3) RQ3: What main themes support the flexibility, scalability, and practical application of microlearning in educational and professional contexts?
- 4) RQ4: What are the challenges and strategic roles emerging from global research, particularly in the context of developing countries?

Thus, the findings of this research contribute to the theoretical development of the microlearning literature and provide practical insights for educators, curriculum

Table 1 Comparison between bibliometric analysis and SLR

Dimension	Bibliometric Analysis	SLR
Purpose	Quantitatively map research output, trends, and networks	Qualitatively synthesize findings and theories across studies
Data Source	Metadata from databases (e.g., Scopus, Web of Science)	Full-text peer-reviewed research articles
Analytical Focus	Citation patterns, co-authorship, keyword co-occurrence, thematic clusters	Study context, methodology, findings, and conceptual contributions
Analytical Tools	Bibliometrix (RStudio), Biblioshiny, VOSviewer, Excel	PRISMA-based coding, thematic synthesis, and critical evaluation
Output	Performance indicators, research maps, thematic evolution	Conceptual insights, research gaps, and practical implications
Level of Analysis	Macro (field- or network-level patterns)	Micro (study-level content and theoretical interpretation)

developers, companies, and policymakers. These insights enhance understanding of the flexibility and scalability of microlearning across diverse educational and professional contexts, including lifelong learning and workplace training, laying the groundwork for continued innovation in digital education and informed design of technology-aligned learning strategies.

This article is systematically structured, beginning with an introduction that presents the concept of microlearning and underscores the significance of this research. The methodology section details the B + SLR approach and outlines the data collection procedures. The results section identifies key trends and influential contributions within the microlearning literature. This is followed by a discussion that not only interprets the findings but also situates them within existing theoretical frameworks and identifies research gaps. The article concludes with a summary of key insights and recommendations. Finally, dedicated sections on limitations and future directions highlight opportunities for further investigation.

Methodology

Research Design and Data Sources

This study employed a hybrid methodology called B + SLR, which combines bibliometric and SLR approaches to examine global trends in microlearning. Bibliometric analysis is a quantitative technique that maps the intellectual structure, research performance, and evolutionary patterns of a scientific field through large-scale analysis of metadata, such as citations, co-authorship, and keyword networks (Donthu et al., 2021). In contrast, an SLR is a theory-oriented and evidence-based approach that systematically synthesizes full-text studies to address predefined research questions and derive conceptual and practical insights (Snyder, 2019). Table 1 summarizes the key differences between the approaches.

By combining these two complementary methods, the B + SLR approach enables both macro-level mapping of research structures and micro-level synthesis of empirical and conceptual insights, thereby providing a holistic understanding of global microlearning research.

The literature selection process strictly adhered to PRISMA guidelines to ensure transparency and rigor throughout the literature selection. The PRISMA framework provides evidence-based standards for reporting systematic reviews and meta-analyses (Moher et al., 2010), thereby promoting consistency and minimizing bias. In this study, the PRISMA process encompassed three key stages: (1) identification, (2) screening (including eligibility assessment), and (3) inclusion (Page et al., 2021). Each

phase was executed with precision to ensure the validity, reliability, and relevance of the selected literature. The detailed workflow is illustrated in the PRISMA flow diagram shown in Fig. 1.

Identification

In the identification phase, literature on microlearning from two reputable sources through August 2024 was searched without strict limitations to ensure broad coverage. The absence of a publication time limit in the search was intended to capture historical and contemporary trends in microlearning research, allowing for the identification and consideration of literature from earlier periods. Scopus and WoS were selected as the primary databases for this study due to their comprehensive coverage of the academic literature (Donthu et al., 2021). A differentiated search strategy was employed: both databases were utilized for query (1) to ensure exhaustive identification of prior review studies, while only Scopus was used for query (2) to capture a substantial, yet manageable, dataset of empirical studies.

Two separate search queries were employed for review and empirical studies. This search combined automated queries with manual validation to ensure that all relevant search terms were adequately accommodated during the identification process. The manual validation involved checking abstracts, keywords, and topic relevance. After the results from both databases were merged, a duplicate elimination process was performed, and the final results were saved for further analysis in the screening stage, as illustrated in Fig. 2.

Review Studies The primary focus of this review is to examine research on microlearning through various types of literature studies, including systematic literature reviews (SLRs), scoping reviews, and bibliometric analyses. The literature search was conducted using the search query (1), which combined terms related to microlearning and review methodologies.

Search query (1).

TITLE-ABS-KEY (("Microlearning" OR "Nanolearning" OR "Bite-sized Learning" OR "Just-in-Time Learning" OR "Learning Nuggets" OR "Chunked Learning") AND ("Bibliometric Analysis" OR "Bibliometric Study" OR "Systematic Literature Review" OR "Scoping Review"))).

According to the search results for review studies, 20 documents were found in Scopus, and 11 documents were found on the WoS, with all identified documents published between 2019 and August 2024. After 11 duplicate documents were eliminated, 20 documents were selected for further analysis by the review studies group. Table 2 summarizes the identification results from this query.

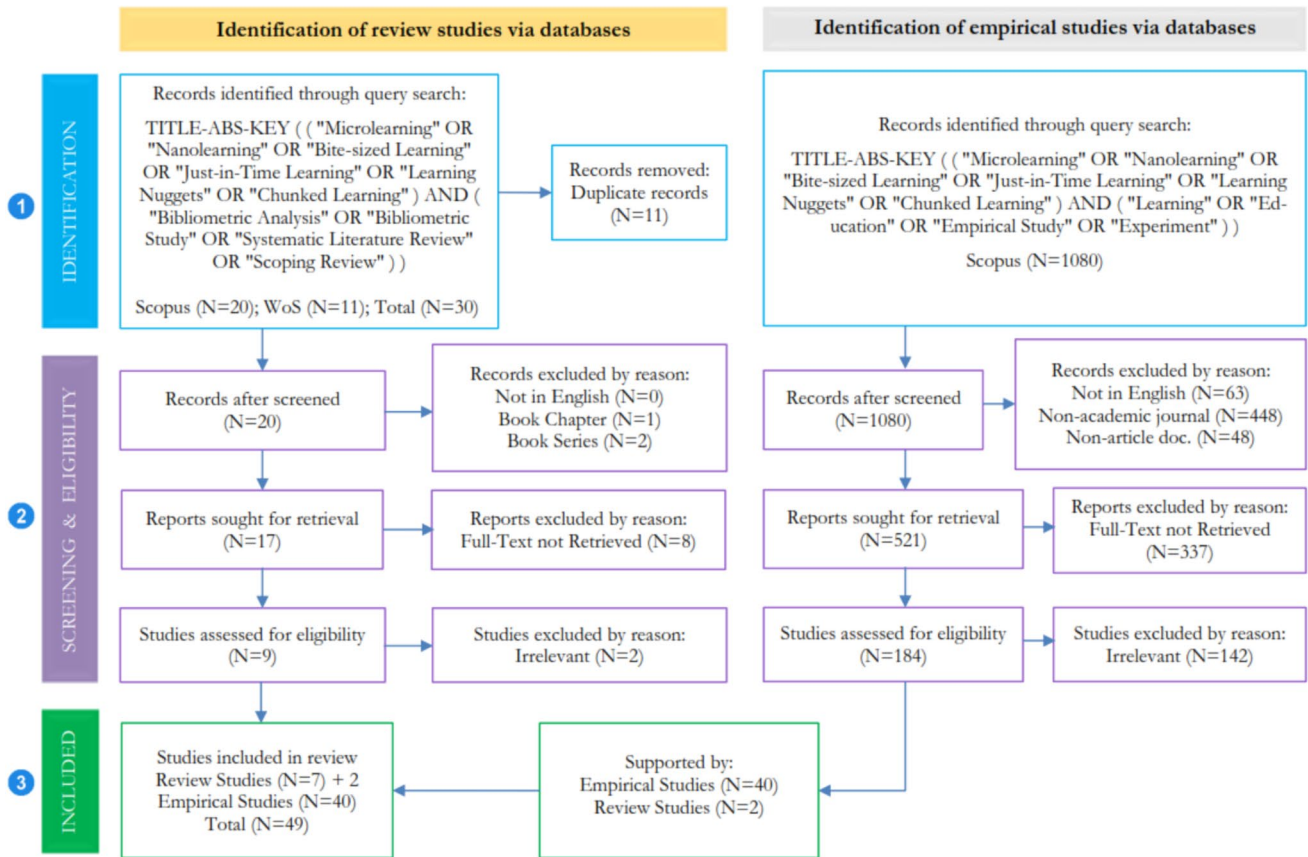


Fig. 1 PRISMA flow diagram

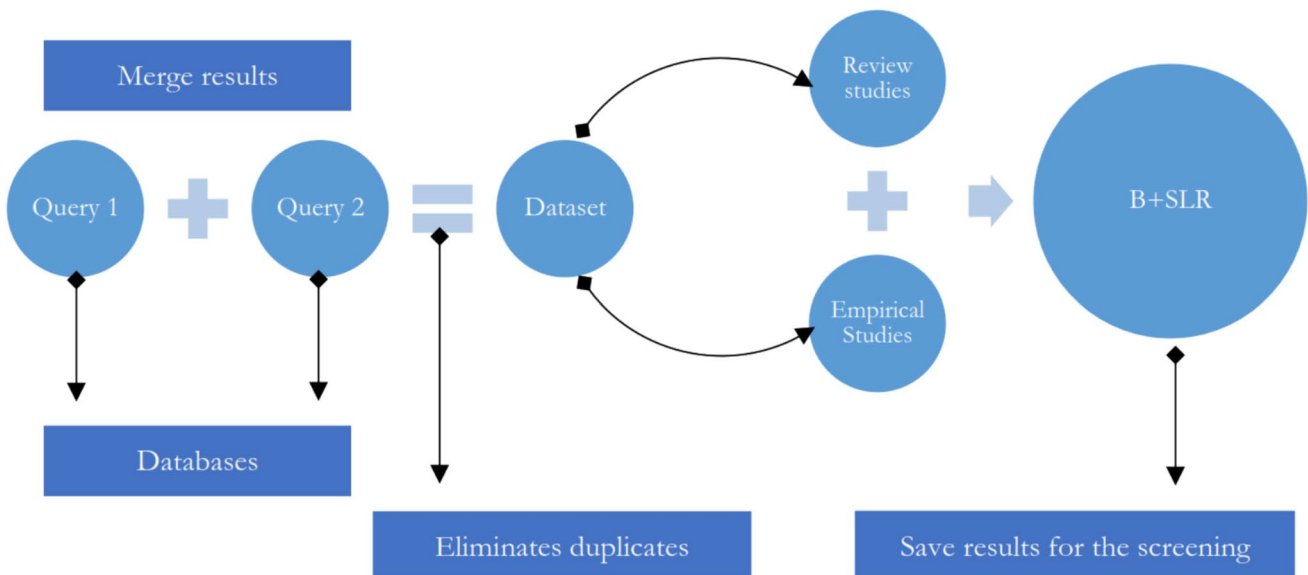


Fig. 2 Identification process in review and empirical studies

Table 2 Summary of review study identification using query (1)

Database	Initial Documents	Time Period	Duplicates	Review Studies
Scopus	20	2019-August 2024	11	20
Web of Science	11	2019-August 2024		

Table 3 Summary of empirical study identification using query (2)

Database	Initial Documents	Time Period	Duplicates	Empirical Studies
Scopus	1,080	1985-August 2024	0	1,080

Empirical Studies Empirical studies contribute essential contextual evidence that complements findings from literature reviews, offering a more comprehensive understanding of microlearning implementation (Budgen et al., 2018; Carrera-Rivera et al., 2022; Gupta et al., 2018). Through methodologies such as experiments, case studies, surveys, and field research, these studies provide direct insights into the practical application, benefits, and limitations of microlearning strategies. This empirical dimension enhances the robustness of this B + SLR by integrating real-world data, thereby offering a more nuanced perspective on the strategic value and challenges associated with deploying microlearning. The corresponding search query (2) was used to retrieve these empirical studies.

Search query (2).

TITLE-ABS-KEY ((“Microlearning” OR “Nanolearning” OR “Bite-sized Learning” OR “Just-in-Time Learning” OR “Learning Nuggets” OR “Chunked Learning”) AND (“Learning” OR “Education” OR “Empirical Study” OR “Experiment”)).

Based on the identification results of the empirical studies, the query was conducted solely in the Scopus database, yielding 1,080 documents published between 1985 and August 2024. This strategic decision was made to ensure the feasibility of the screening process. While search query (1) utilized both databases to capture all prior reviews, the 1,080 documents identified from Scopus already provide a large and robust foundation for a B + SLR. The substantial number of documents highlights the growing scholarly interest in microlearning over the past decades. The next step involves filtering these documents to ensure that the studies selected for further analysis are both relevant and of high quality. Table 3 presents a summary of the identification results.

Data Screening and Eligibility

Following the identification phase, the screening process was implemented to ensure the relevance and quality of the retrieved documents. This process categorized the studies into two main groups: (a) Review Articles and (b) Empirical Studies, each subject to specific inclusion and exclusion criteria. The screening and eligibility phases aimed to retain only those publications demonstrating strong relevance to the topic and sufficient scholarly rigor. During the eligibility assessment, articles were further examined for methodological robustness, reliability of data, and the extent of their theoretical or empirical contribution to microlearning research. Publications that referred to microlearning only tangentially, without offering substantive analysis or discussion, were excluded. This rigorous filtering process is essential to guarantee that the final dataset yields valid and meaningful insights (Donthu et al., 2021; Lim et al., 2024).

Eligibility for Review Studies Regarding the criteria for review studies, only papers with a methodology that adhered to systematic review or bibliometric analysis were included. English-language and open-access articles were used as the selection criteria for the papers to ensure consistency in quality and accessibility. English was chosen because it is the most commonly used language in academic literature, facilitating comparative evaluations at the international level. Open-access selection helps mitigate accessibility bias, ensuring that all the analyzed literature is widely accessible to researchers without barriers. Documents meeting the inclusion criteria were ready for further analysis.

Exclusion Criteria:

- 1) Document Type: Non-article documents (e.g., reports, book reviews).
- 2) Language: Non-English articles.
- 3) Source Type: Non-academic journal publications.
- 4) Access Type: Non-open-access articles.
- 5) Publication Year: Articles published after August 2024 were excluded due to the cutoff of the data collection period.

Inclusion Criteria:

- 1) Document Type: Only articles, reviews, and conference proceedings are accepted.
- 2) Language: Must be written in English.
- 3) Source Type: Only journals and conference proceedings are considered.
- 4) Access Type: Only open-access articles are included.
- 5) Publication Year: No time limit; all articles published up to August 2024 were included.
- 4) Access Type: Only open-access articles are included.
- 5) Keywords: Must include terms such as microlearning, just-in-time learning, and bite-sized learning.
- 6) Publication Year: There is no limit on publication year; all articles up to August 2024 are included.

A total of 11 documents were excluded for failing to meet the established inclusion criteria, including non-article formats, non-academic publications, or a lack of open access availability. After this screening process, nine review studies that fully satisfied the inclusion criteria were retained for further analysis. Only those studies employing explicit methodologies and adhering to recognized systematic review or bibliometric analysis standards were considered. The selected review studies spanned publications from 2019 through August 2024.

Eligibility for Empirical Studies In the screening process for empirical studies, the focus is on articles that contain empirical research related to microlearning. Documents are evaluated based on several criteria: (1) relevance to microlearning or just-in-time learning, (2) use of a clear and robust methodology, such as experiments, case studies, or surveys, and (3) significant theoretical or empirical contributions. Articles that merely mentioned microlearning superficially, without in-depth analysis, were excluded from the analysis.

A different source type criterion was applied for this group. While conference proceedings were included for review studies to capture emerging trends, they were excluded from the empirical studies group. This decision was made to ensure that only empirical data that had passed through the full peer-review process of academic journals was included in the analysis.

Exclusion Criteria:

- 1) Document Type: Non-article documents (e.g., reports, book reviews).
- 2) Language: Non-English documents.
- 3) Source Type: Non-academic journals.
- 4) Access Type: Non-open-access articles.
- 5) Keywords: Missing relevant keywords like “microlearning” or “just-in-time learning.”
- 6) Publication Year: Articles published after August 2024.

Inclusion Criteria:

- 1) Document Type: Only articles.
- 2) Language: Must be written in English.
- 3) Source Type: Only journals.

Potential biases impacting the results were carefully considered. Publication bias, which often favors positive or significant findings in high-impact journals, was mitigated by including a broad range of sources without restricting the search to high-impact journals and by incorporating open-access articles to capture diverse perspectives. Consequently, 896 documents were excluded for not meeting the inclusion criteria, including non-article document types (48), non-academic publications (448), non-English language articles (63), and articles lacking open access (337). Following this rigorous screening, 184 documents fulfilling all inclusion criteria were retained for subsequent analysis within the empirical studies group.

Following the initial screening, a rigorous eligibility assessment was performed to confirm the relevance of articles to microlearning research. Within the review studies category, 2 articles were excluded due to irrelevance, resulting in 7 eligible documents. In the empirical studies category, 142 articles were similarly excluded for lack of relevance. Notably, two articles, Chamorro-Atalaya et al. (2024) and Leong et al. (2021), originally categorized as empirical studies, were reclassified as review studies owing to their substantial contributions via comprehensive systematic reviews. Consequently, a total of 49 documents were included in the final analysis, comprising 9 review studies and 40 empirical studies.

It is important to note that the B + SLR framework fundamentally differs from traditional large-scale bibliometric analyses. Rather than mapping thousands of publications, our approach deliberately filters the corpus to the 49 most thematically relevant and high-impact studies, in accordance with our PRISMA inclusion criteria. Consequently, the subsequent bibliometric analysis is focused on mapping the elite landscape of these core articles, rather than the entire field.

Documents Included in the Analysis

After the screening and eligibility processes, 49 documents meeting the inclusion criteria were selected for further analysis, comprising 9 review studies and 40 empirical studies. These documents were filtered based on document type, language, accessibility, and relevance to microlearning. Figure 1 (PRISMA flow diagram) shows the selection process. The datasets were combined into a CSV file for bibliometric analysis using RStudio and Bibliometrix. This hybrid approach integrates both empirical and systematic

literature, providing a comprehensive perspective on microlearning trends.

Quality and Credibility Assurance

The data collection process in this study was conducted rigorously to ensure that only articles meeting quality criteria were included in the analysis. Each bibliographic metadata entry was carefully reviewed to ensure the accuracy and completeness of the information collected from each article. This verification process included checking key details such as title, author, publication year, keywords, and citations, which are essential for maintaining the data quality used in the bibliometric analysis. This ensures that the study's results are reliable and valid.

The results from Table 4 show that most metadata elements, such as abstracts (AB), author affiliations (C1), DOI (DI), total citations (TC), and titles (TI), were 100% complete. This approach is a strong foundation for bibliometric research because it ensures that the core components of each analyzed article are available. With the completeness of these elements, analyses such as citation analysis and author impact analysis can be conducted with high accuracy, as the complete data allows for a comprehensive mapping of citation patterns and the impact of authors or journals.

Despite the high completion rate of most metadata elements, some notable gaps may impact the overall comprehensiveness of the analysis. The keyword (DE) and publication year (PY) fields showed a 98% completion rate, indicating minimal missing data. However, gaps in keyword data can affect the accuracy of thematic and co-occurrence analyses, which are essential for identifying key research trends. The corresponding author (RP) metadata field is 82%

complete, which may limit the depth of author collaboration network analysis, an important component for mapping institutional and researcher relationships in microlearning studies. In contrast, metadata fields such as Keywords Plus (ID), Cited References (CR), and Science Categories (WC) exhibited 100% data loss, which is critical, as it significantly constrains citation network construction and thematic exploration, thereby hindering the identification of broader scholarly linkages and core research themes.

Although metadata elements such as Keywords Plus (ID), Cited References (CR), and Science Categories (WC) are entirely missing, this deficiency can be effectively mitigated by leveraging the author-supplied keywords (DE), which demonstrated a 98% completion rate. While this omission limits certain dimensions of citation network and thematic analyses, the high completeness of the DE data still enables a robust keyword co-occurrence analysis to map core research themes (van Eck & Waltman, 2010; Zhang et al., 2016). Therefore, despite these specific data gaps, the high level of completeness across other essential metadata ensures the study proceeds with a reliable foundation, allowing for the extraction of meaningful insights and a substantial contribution to the field (Aria & Cuccurullo, 2017; Moed, 2005).

Data Analysis

This study utilized RStudio with the Bibliometrix package to perform quantitative analyses and generate visualizations of bibliometric networks within the microlearning literature. The analysis began with keyword frequency analysis to identify the most commonly occurring terms across the selected articles. Further analyses included mapping citation networks and exploring collaboration patterns among

Table 4 Completeness of bibliographic metadata

Metadata	Description	Missing Counts	Missing %	Status
AB	Abstract	0	0.00	Excellent
C1	Affiliation	0	0.00	Excellent
AU	Author	0	0.00	Excellent
DI	DOI	0	0.00	Excellent
DT	Document Type	0	0.00	Excellent
SO	Journal	0	0.00	Excellent
LA	Language	0	0.00	Excellent
TI	Title	0	0.00	Excellent
TC	Total Citation	0	0.00	Excellent
DE	Keywords	1	2.00	Good
PY	Publication Year	0	0.00	Excellent
RP	Corresponding Author	9	18.00	Acceptable
ID	Keywords Plus	38	76.00	Critical
CR	Cited References	50	100.00	Completely missing
WC	Science Categories	50	100.00	Completely missing

authors and institutions (Aria & Cuccurullo, 2017; Donthu et al., 2021). This analysis is critical for uncovering key themes, conceptual frameworks, and evolving areas of focus in microlearning research. The main tools utilized in the data analysis process are described as follows:

- 1) **RStudio with the Bibliometrix Package:** This software is the primary tool for conducting in-depth bibliometric analysis (Aria & Cuccurullo, 2017). It is utilized for citation network analysis and research trend mapping, generating informative data visualizations. The Biblioshiny interface (<https://www.bibliometrix.org/>) enhances dataset management, making it interactive and user-friendly. A significant advantage of RStudio with Bibliometrix is its ability to integrate datasets from multiple sources, such as Scopus and WoS, facilitating comprehensive analysis. Its flexibility in handling co-citation analysis, keyword co-occurrence, and dynamic thematic mapping makes Bibliometrix highly suitable for this research.
- 2) **Microsoft Excel:** This tool plays a crucial role in the data filtering and organization stages before further analysis with other software is conducted. Beyond data management, Excel is also effective for visualization, allowing for the creation of clear and comprehensible tables, graphs, and charts. Its built-in formulas support quick calculations, data aggregation, and the development of basic visualizations, providing a preliminary overview before in-depth analysis (Donthu et al., 2021; Guleria & Kaur, 2021).
- 3) **Python:** This programming language is essential for data processing, quantitative analysis, and complex visualization. With libraries such as Pandas, Python efficiently manages large bibliometric datasets. The Matplotlib and Seaborn libraries were used for detailed data visualization. Additionally, statistical analysis was used to calculate word frequency, citation counts per article, and bibliometric indicators such as the h-index

and g-index (Haustein & Larivière, 2015; van Eck & Waltman, 2014).

Citation analysis was performed to identify citation patterns and measure the impact of specific articles (Moed, 2005; Small, 1973). This approach helps researchers understand how frequently an article is cited and how it influences microlearning development. Additionally, influential documents in higher education and corporate training can be identified through this approach. Productivity analysis was also conducted to evaluate the contributions of individual authors and institutions (van Eck & Waltman, 2014). This analysis highlights the most productive researchers and institutions shaping the microlearning literature.

Additionally, a comprehensive trend analysis was conducted to trace the evolution of microlearning research over time, elucidating shifts in research focus and the expanding applications of microlearning in education and training. Notably, this analysis was designed to identify dominant themes, such as flexibility and scalability, and to map their development in contemporary learning environments (Samala et al., 2023a, 2023b). Network visualization techniques were employed to map collaborations among researchers and institutions, revealing the interconnected nature of microlearning research and emphasizing international partnerships. Utilizing RStudio, collaboration networks were clearly visualized, providing valuable insights into global cooperative trends. By integrating these methodologies, this study effectively addresses critical questions regarding global research trends, leading contributors, and key thematic areas that reinforce microlearning's role in fostering flexible and scalable learning solutions across educational and professional domains (Figs. 3, 4 and 5).

Fig. 3 Results of demographic distribution



Fig. 4 Annual publication trends (included subscription docs.)

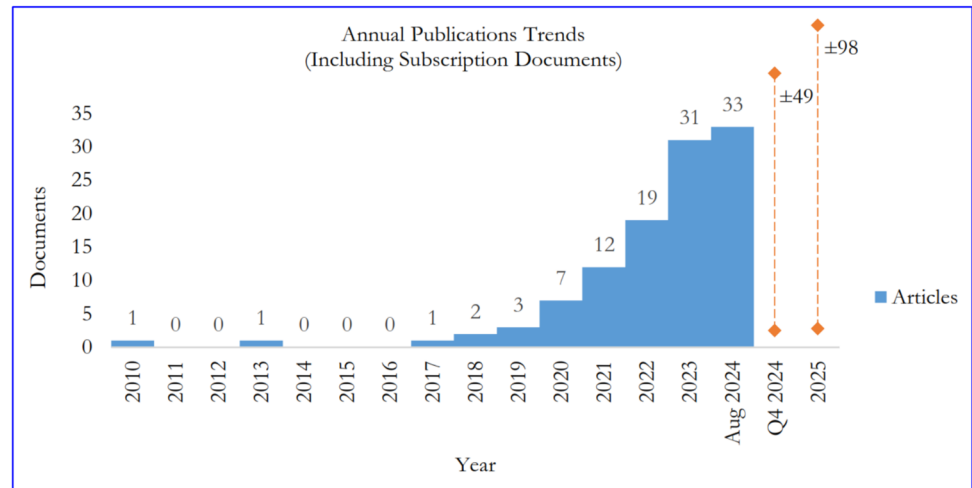
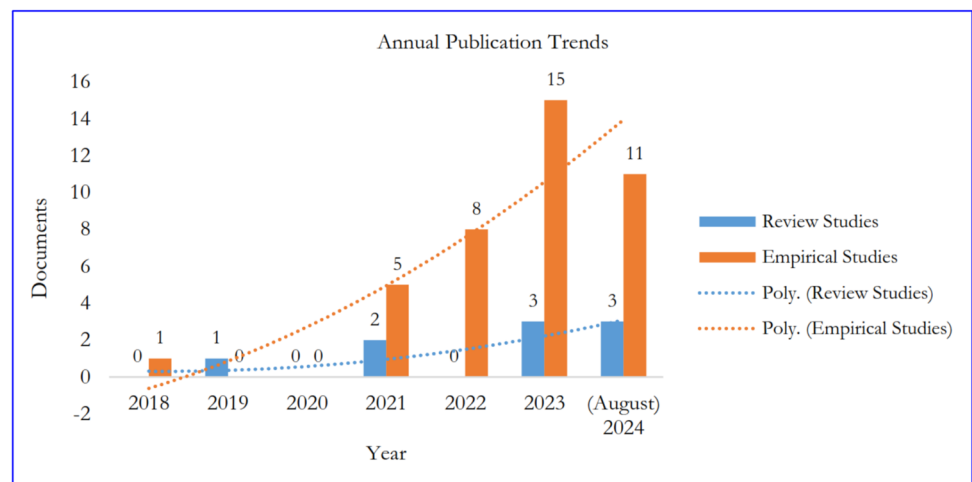


Fig. 5 Annual publication trends



Results

Results of Demographic Distribution

This study analyzed 49 microlearning research articles published between 2018 and August 2024. As illustrated in Fig. 6, the annual growth rate reached a remarkable 55.25%, demonstrating a sharp increase in research interest and underscoring microlearning's growing relevance in addressing the challenges posed by the digital era. This surge coincides with the widespread adoption of digital technologies and the rising trend toward flexible learning modalities. The COVID-19 pandemic notably accelerated this momentum by amplifying demand for remote learning solutions. A total of 169 authors contributed to these publications, with 7 articles authored individually. The rate of international collaboration is substantial, with 20.41% of papers featuring co-authors from multiple countries, indicating that microlearning research transcends localized

interest and enjoys global recognition as an effective educational and training methodology across diverse settings. On average, each article involved 3.59 authors, reflecting the collaborative nature of the field. The analyzed literature incorporated 173 unique keywords, reflecting the broad thematic and conceptual diversity within microlearning research, spanning technology, pedagogy, and various application contexts. The average document age was 1.35 years, highlighting the currency and timeliness of the field's scholarship. Furthermore, with an average citation count of 9.12 per document, microlearning research demonstrates significant academic influence.

The fact that all the analyzed documents fall within this timeframe highlights two key observations. First, microlearning is still in its rapid growth phase, with many recent publications forming the foundation of this research. Second, although there is older literature, the most significant contributions to the field's development occurred between 2018 and 2024, underscoring the relevance of microlearning to evolving modern learning trends and needs. Figure 3

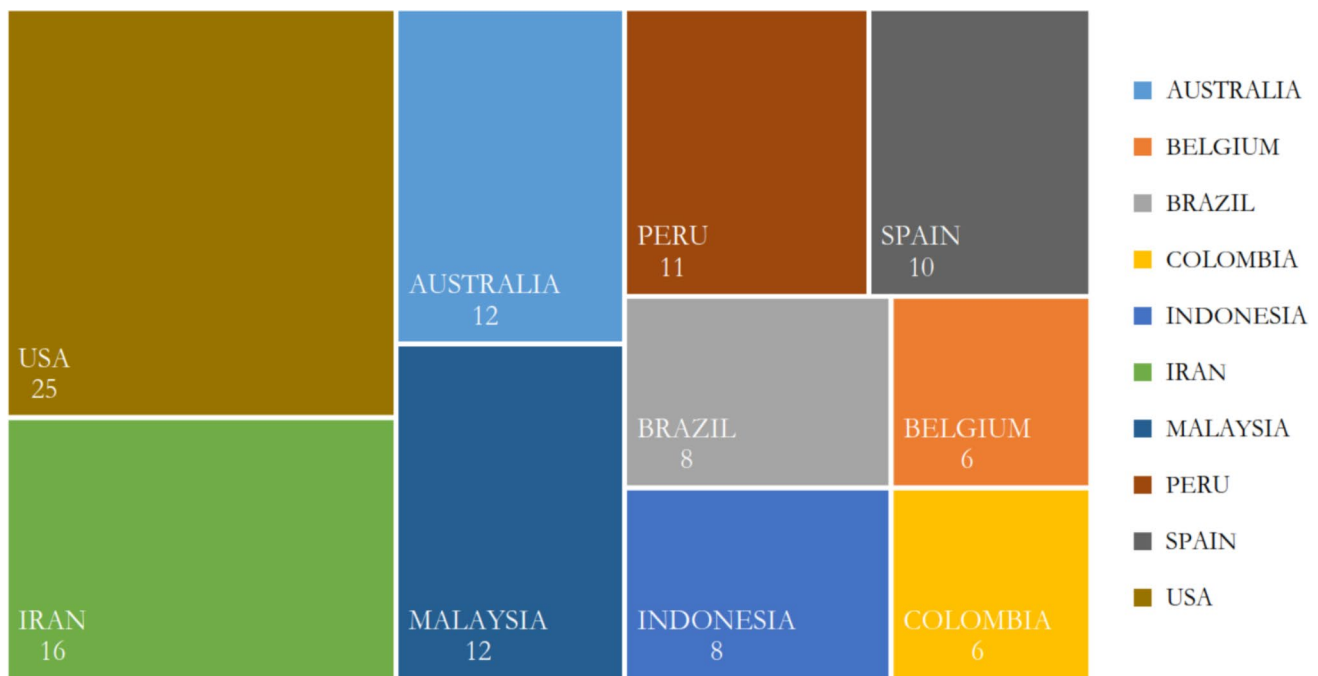


Fig. 6 Top 10 countries in microlearning research (2018–August 2024)

shows the demographic information regarding the final dataset.

Global Trends in Microlearning

Annual Publication Trends

Figure 4 illustrates a slow growth in microlearning-related publications from 2010 through August 2024, with only three articles published in 2010, 2013, and 2017 respectively, indicating limited early research interest. However, this trend shifted beginning in 2018, with two publications that year, followed by a gradual increase to three articles in 2019. A notable surge occurred in 2020, with seven publications, likely driven by the widespread adoption of educational technologies and the urgent demand for distance learning solutions amid the COVID-19 pandemic.

By 2021, the number of publications rose significantly, with 12 articles, followed by 19 in 2022, and reached 31 articles in 2023, signaling that a growing interest in microlearning as an educational technology continued to expand across multiple sectors. By August 2024, 33 articles had been published, demonstrating a continuous growth trend. In contrast, the open-access dataset reveals similar trends, with a noticeable increase in publications from 2018 to August 2024. Initial publications in 2018 included only one empirical study, with no review studies. By 2019, one review study was published, but no empirical studies were available.

Publication numbers stagnated in 2020, likely due to the global impact of COVID-19.

Starting in 2021, there was a significant increase in microlearning-related publications, with two review studies and five empirical studies. This rise can be attributed to the rapid digitalization prompted by the COVID-19 pandemic, which encouraged institutions and companies to adopt flexible, remote learning solutions such as microlearning. The upward trend continued in 2022, with eight empirical studies published, followed by the largest jump in 2023, which saw three review studies and fifteen empirical studies. This surge reflects the widespread adoption of microlearning, particularly in the corporate and higher education sectors (Borrego Ramírez & Ruiz Cansino, 2023; Corbeil et al., 2023; Robles et al., 2023; Romanenko et al., 2023; Shabadurai et al., 2022).

Educational institutions began incorporating microlearning to provide flexible learning options, while companies used it for quick training to enhance employee skills. Empirical studies have increasingly been conducted to assess the effectiveness of microlearning in real-world applications, from classroom instruction to decentralized work environments. As of August 2024, three review studies (Chamorro-Atalaya et al., 2024; Fialho et al., 2024; Pham et al., 2024) and 11 empirical studies (Alshammari, 2024; Al-Zahrani, 2024; Aouad et al., 2024; Auf, 2024; Balasundaram et al., 2024; Casebourne, 2024; Conde-Caballero et al., 2024; Lykova & Savelyeva, 2024; Naser, 2024; Rahbar et al., 2024; Rof et al., 2024) have been published. Although data for

the remainder of 2024 are not yet available, the trend suggests that interest in microlearning continues to grow, supported by technological advancements and the increasing demand for flexible learning across educational and corporate sectors.

Figure 5 highlights how publication trends, particularly empirical studies, have shown significant growth since 2021, peaking in 2023. This increase also indicates that microlearning is increasingly recognized as a relevant and effective approach. Among the publication trends in the field of microlearning, there is a distinct difference in impact between review studies and empirical studies. Review studies offer a comprehensive overview of the literature, synthesizing previous research findings to provide deeper theoretical insights. These findings can help researchers understand trends, gaps, and future directions in microlearning research.

Conversely, empirical studies provide concrete evidence of the practical implementation of microlearning. An increasing number of empirical studies, especially post-2021, indicate a growing interest in exploring the practical applications of microlearning in sectors such as higher education and corporate training. These studies offer direct insights into the effectiveness and efficiency of microlearning in enhancing learning outcomes through classroom experiments or workplace simulations. The overall trend clearly shows an increase in microlearning-related publications, signifying its growing recognition as an effective learning method across educational and industrial contexts.

Country-Level Analysis

Country-level analysis reveals that the United States (USA) led in terms of microlearning research publications with 25 documents published between 2018 and August 2024. The influence of the USA is further emphasized by a high citation count of 166, averaging 33.20 citations per article. This position of the USA as a primary hub for global microlearning

research is bolstered by strong research infrastructure and collaboration with universities and technology companies.

Notably, Iran and Malaysia also contributed 16 and 12 documents, respectively, indicating that developing countries are increasingly involved in microlearning development. However, their citation impact remains relatively lower than that of developed nations, suggesting that while publication numbers are rising in these countries, their global research influence is still limited, as depicted in Fig. 6.

Table 5 provides detailed information in terms of total citations and average article citations from the country's perspective. The USA published 25 papers and has been cited 166 times. European nations like Spain and the UK also contribute to microlearning research. Spain has published 10 documents with 25 citations, while the UK, despite having only 3 publications, recorded 81 citations, averaging 27 citations per article. This suggests that UK research, although limited in volume, has a considerable impact.

Countries such as Indonesia and Brazil have also significantly contributed to the field, each publishing 8 documents. Notably, Indonesia has garnered 25 citations, averaging 12.50 citations per article, positioning it among developing nations and gaining global academic recognition. This indicates that although Indonesia's publication volume is smaller than that of the United States, the quality and relevance of its research are increasingly acknowledged. Conversely, Brazil's 8 documents have not yet achieved significant citation impact, reflecting a lack of international recognition.

The growth in Indonesia and Brazil indicates rising interest and international collaboration. Indonesia's strong performance reflects growing global acknowledgment, while countries like Spain and Belgium demonstrate high levels of cross-national collaboration, enhancing their research impact. However, the gap between developed and developing countries remains evident, with the USA and the UK dominating citation impact and global relevance. It is also important to note that some countries may focus more on the application of microlearning in specific sectors, such as higher education and corporate training. For instance, Brazil and Indonesia appear to invest more in research related to implementing microlearning in the education sector, whereas the USA tends to adopt a broader approach encompassing various industry sectors.

International collaboration is also an important factor in the country-level analysis. As shown in Table 6, the United States has 4 single-country publications (SCP) and only 1 international collaboration (MCP), indicating that most of the research in the country has been conducted independently. In contrast, countries such as the United Kingdom and Spain have higher levels of international collaboration, with 66.7% of MCPs in the United Kingdom and 33.3% in Spain. This active collaboration may explain their significant impact on the global academic community.

Table 5 Most cited countries

No	Country	Total Citations	Average Article Citations
1	USA	166	33.20
2	UNITED KINGDOM	81	27.00
3	INDONESIA	25	12.50
4	SPAIN	25	8.30
5	TURKEY	21	21.00
6	IRAN	16	5.30
7	CHINA	11	11.00
8	CANADA	8	8.00
9	AUSTRALIA	7	3.50
10	AUSTRIA	6	6.00

Table 6 Most relevant countries

No	Country	Articles	Articles %	SCP	MCP	MCP %
1	USA	5	10.2	4	1	20
2	IRAN	3	6.1	3	0	0
3	SPAIN	3	6.1	2	1	33.3
4	UNITED KINGDOM	3	6.1	1	2	66.7
5	AUSTRALIA	2	4.1	2	0	0
6	BELGIUM	2	4.1	0	2	100
7	INDONESIA	2	4.1	1	1	50
8	MALAYSIA	2	4.1	2	0	0
9	SAUDI ARABIA	2	4.1	2	0	0
10	AUSTRIA	1	2	1	0	0

Table 7 Top keywords in microlearning research (2018 – August 2024)

Words	Occurrences
microlearning	42
e-learning	8
higher education	7
online learning	4
technology acceptance	3
bite-sized learning	2
distance learning	2
flipped classroom	2
knowledge	2
learning	2
learning analytics	2
mobile application	2
multimedia	2
self-efficacy	2
social media	2
video	2
2-mode analysis	1
21st-century learning	1
active learning	1

With an MCP of 50%, Indonesia demonstrates substantial potential for expanding international collaboration, which could significantly enhance its global research impact. In contrast, Belgium's MCP of 100% indicates that all its microlearning publications result from cross-country partnerships, underscoring the critical role of international collaboration in amplifying research relevance and global reach. While developing countries such as Indonesia, Malaysia, and Iran are beginning to increase their contributions to microlearning research, advanced countries such as the United States, the United Kingdom, and Spain continue to lead in both volume and quality. This trend indicates the potential for further growth, especially with international collaboration and the strengthening of the educational technology

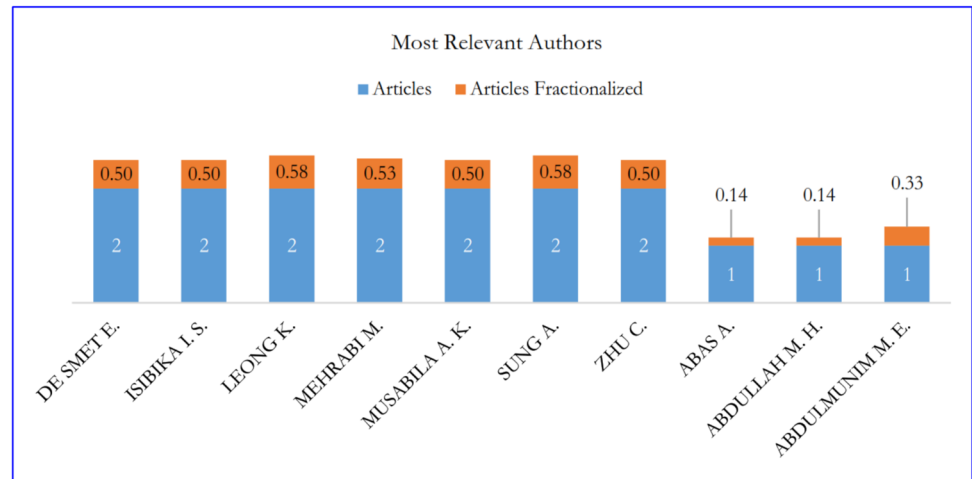
sector in developing countries. However, greater investment in research and infrastructure in these countries is needed to support the equitable development of microlearning research in the future.

Keyword Analysis

As shown in Table 7, the most frequently occurring keyword was microlearning, which appeared 40 times. This emphasizes its central theme across all the analyzed documents. A high frequency indicates a consistent focus on learning through small, easily digestible modules throughout the research field. This pattern highlights the importance of microlearning in contemporary educational strategies, reinforcing its relevance in various contexts.

With 42 occurrences, the keyword microlearning underscores the central focus on delivering educational content in small, manageable units to enhance retention and learner engagement. Related keywords such as e-learning (8 occurrences) and higher education (7 occurrences) highlight the strong association between microlearning and contemporary educational technologies, particularly within online learning environments. The growing prominence of e-learning reflects its integral role in digital educational strategies, especially in higher education, where flexible learning models are increasingly critical. This trend emphasizes the seamless integration of microlearning with digital platforms to enrich educational experiences, notably in virtual settings.

A substantial proportion of studies on microlearning focus on college and university contexts, aiming to develop accessible, scalable, and student-centered instructional approaches. Emerging keywords such as bite-sized learning (2 occurrences) and learning analytics (2 occurrences) point to growing interest in modular content delivery and data-driven optimization of teaching methodologies. Likewise, the presence of self-efficacy (2 occurrences) reflects a deeper focus on learners' confidence and belief in their ability to succeed within microlearning frameworks, aligning

Fig. 7 Most relevant authors in microlearning research (2018–August 2024)**Table 8** Author productivity through lotka's law

Documents written	N. of Authors	Proportion of Authors
1	163	0.959
2	7	0.041

with current discussions on personalized and motivational learning design.

Key Contributors and Influential Institutions

Most Relevant Authors

By analyzing the most contributing authors, we identified several scholars who consistently published microlearning articles between 2018 and August 2024. Figure 7 presents a list of the most relevant authors and the number of published articles. Notable authors, such as De Smet E., Isibika I.S., and Leong K., stand out for their consistent contributions to this field. The “fractionalized articles” metric reflects the number of articles contributed by each author

in collaborative research, providing insight into the relative contribution of each author across various studies.

Author Productivity Through Lotka's Law

Utilizing Lotka's Law, commonly employed in bibliometrics to model author productivity distribution, we analyzed authors' productivity in microlearning. The results indicate that most of the authors have written only a single article, while a small fraction have contributed two articles. In this study, we found that 95.9% of the authors contributed to only one document, with only 4.1% publishing more than one article in microlearning research, as presented in Table 8.

Author Impact: h-index, g-index, and m-index

In addition to assessing the publication count, we evaluated the impact of each author using the h-index, g-index, and m-index. These indices help measure the quality and overall impact of an author's contributions. Authors such as Leong K. and Sung A. stand out with h-indexes of 2 and total citations (TCs) of 80, indicating that their research has had a significant influence on the microlearning literature. It

Table 9 Author impact

No	Author	h_index	g_index	m_index	TC	NP	PY_start
1	LEONG K	2	2	0.500	80	2	2021
2	SUNG A	2	2	0.500	80	2	2021
3	AU D	1	1	0.250	73	1	2021
4	ABRAMENKA-LACHHEB V	1	1	0.500	20	1	2023
5	BALLESTEROS-YÁNEZ I	1	1	1.000	13	1	2024
6	ALIAS N.F	1	1	0.500	4	1	2023
7	ABAS A	1	1	0.500	2	1	2023
8	ABDULLAH M.H	1	1	0.500	2	1	2023
9	ALARCON R.T	1	1	0.333	1	1	2022
10	BANNACH G	1	1	0.333	1	1	2022

Table 10 Most relevant affiliations

No	Affiliation	Articles
1	Shiraz University of Medical Sciences (IRAN)	8
2	Tehran University of Medical Sciences (IRAN)	8
3	InsideOut Institute (AUSTRALIA)	7
4	Universiti Pendidikan Sultan Idris (MALAYSIA)	7
5	Universidad del Norte (COLOMBIA)	6
6	Universidad Nacional Tecnológica de Lima Sur (PERU)	5
7	Universidade Estadual Paulista (BRAZIL)	5
8	Universitas Pendidikan Indonesia (INDONESIA)	5
9	Vanderbilt University (USA)	5
10	Nanyang Technological University (SINGAPORE)	4

is notable that Leong K. and Sung A. collaborated on both published documents, which have garnered significant citations, reflecting the strength of their combined contributions to the microlearning literature. Their relatively high and consistent citation counts, as reflected in Table 9, demonstrate the notable impact of their collaborative work on the field.

Most Influential Institutions

Table 10 highlights several institutions that have emerged as key players in microlearning research from 2018 to August 2024. The data reveal that Shiraz University of Medical Sciences and Tehran University of Medical Sciences each lead with 8 articles, as depicted in Table 10. The dominance of these Iranian institutions suggests a growing emphasis on microlearning in health education, particularly in medical training contexts.

Additionally, the InsideOut Institute from Australia and the Universiti Pendidikan Sultan Idris from Malaysia also emerged as significant contributors, each with 7 articles. Institutions from the Asia–Pacific region, particularly Universiti Pendidikan Sultan Idris, exemplify the growing implementation of microlearning in formal education contexts, especially in developing countries such as Malaysia.

Vanderbilt University in the USA and Nanyang Technological University in Singapore also significantly contributed with 5 articles each. The impact of institutions from the USA is not surprising given their substantial contributions to global research; however, the involvement of institutions from Singapore indicates that Southeast Asia is beginning to play a larger role in this area of research, aligning with the growth of the edtech sector in the region.

Overall, the contributions from these institutions indicate that while North America and Europe still dominate global research, other regions, such as Asia and the Middle East, are demonstrating a significant increase in influence, particularly in technology-based education and health sectors.

Table 11 Collaboration networks

From	To	Frequency
BELGIUM	TANZANIA	2
UNITED KINGDOM	HONG KONG	2
CANADA	PAKISTAN	1
INDONESIA	SERBIA	1
INDONESIA	TÜRKIYE	1
SPAIN	MEXICO	1
TÜRKIYE	SERBIA	1
USA	CANADA	1
USA	KOREA	1
USA	QATAR	1

Collaboration Networks

International collaboration plays a vital role in advancing microlearning research. The collaboration network analysis revealed several noteworthy relationships among countries engaged in joint publications. For example, Belgium and Tanzania recorded two instances of collaboration, suggesting an emerging bilateral research partnership within this field. Indonesia has also engaged in collaborative efforts with Serbia and Türkiye, reflecting growing regional cooperation in educational technology and digital learning. Notable partnerships include those between the United States and Canada, the United States and Korea, and the United States and Qatar. Although the number of U.S. collaborations is limited, these connections indicate its strategic engagement with both neighboring and distant regions, particularly across North America, Asia, and the Middle East. Overall, the current collaboration patterns suggest that while international research cooperation in microlearning is developing, it remains relatively concentrated and has yet to expand into a truly global network (Table 11).

According to the node-cluster data regarding author collaboration, authors such as Leong K. and Sung A. from Cluster 1 play central roles in microlearning research, as indicated by their PageRank scores, placing them at the heart of collaboration efforts. Authors such as De Smet E. and Musabila A.K. from Cluster 2 also exhibited significant influence, albeit within smaller networks.

Tables 12 and 13 provide detailed insights into key research contributions in the field of microlearning, including empirical studies and literature reviews. Table 12 lists the articles ranked by total citations received, offering insights into the most influential contributions driving global microlearning adoption from 2018 to August 2024. Conversely, Table 13 summarizes various systematic reviews conducted by researchers, covering methodologies, key findings, and limitations faced in their studies. Together, these tables serve as essential tools for understanding the evolution of

Table 12 List of articles by total citations (TC) and research areas in microlearning (2018– August 2024)

No	Paper	Study	Title	TC	Research Areas
1	De Gagne et al. (2019a)	Review	Microlearning in Health Professions Education: Scoping Review	120	Health Professions Education
2	Leong et al. (2021)	Review	A Review of the Trend of Microlearning	73	General Microlearning, Review
3	Park and Kim (2018)	Empirical	A Design and Development of Microlearning Content in an E-Learning System	29	e-Learning, Instructional Design
4	Fidan (2023)	Empirical	The Effects of Microlearning-Supported Flipped Classroom on Pre-Service Teachers' Learning Performance, Motivation, and Engagement	21	Teacher Education, Flipped Classroom
5	Sankaranarayanan et al. (2023)	Review	Microlearning in Diverse Contexts: A Bibliometric Analysis	20	Bibliometric Analysis, General
6	Samala et al., (2023a, 2023b)	Review	Microlearning: Transforming Education with Bite-Sized Learning on the Go—Insights and Applications	16	Higher Education, Mobile Learning
7	McNeill and Fitch (2023)	Empirical	Microlearning through the Lens of Gagne's Nine Events of Instruction: A Qualitative Study	14	Instructional Design, Higher Education
8	Zarshenas et al. (2022)	Empirical	The Effect of Microlearning on Learning and Self-Efficacy of Nursing Students: An Interventional Study	14	Health Education, Nursing
9	Conde-Caballero et al. (2024)	Empirical	Microlearning through TikTok in Higher Education: An Evaluation of Uses and Potentials	13	Higher Education, Social Media
10	Romero-Rodríguez et al. (2023)	Empirical	Collaborative Online International Learning between Spain and Mexico: A Microlearning Experience to Enhance Creativity in Complexity	12	Collaborative Learning, Higher Education
11	Zheng (2021)	Empirical	Research on Data Retrieval Algorithm of English Microlearning Teaching Based on Wireless Network Information Classification	11	Language Learning, Wireless Networks
12	Triana et al. (2021)	Empirical	Quiz-Based Microlearning at Scale: A Rapid Educational Response to COVID-19	10	Medical Education, Pandemic Response
13	Susilana et al. (2022)	Empirical	Can Microlearning Strategy Assist Students' Online Learning?	9	Online Learning, Educational Technology
14	Yilmaz et al. (2022)	Empirical	The Learning Loop: Conceptualizing Just-in-Time Faculty Development	8	Faculty Development, Just-in-Time Learning
15	Sung et al. (2023)	Empirical	A Study of Learners' Interactive Preference on Multimedia Microlearning	7	Multimedia Learning, Interactivity
16	Garshasbi et al. (2021)	Empirical	Microlearning and Computer-Supported Collaborative Learning: An Agenda Towards a Comprehensive Online Learning System	7	Collaborative Learning, Online System
17	Shabadurai et al. (2022)	Empirical	Investigating the Employees' Perspectives and Experiences of Microlearning Content Design for Online Training	6	Corporate Training, Content Design
18	Mujica-Luna et al. (2021)	Empirical	Microlearning Platforms Brand Awareness Using Social Media Marketing and Customer Brand Engagement	6	Social Media, Marketing, Brand Engagement

Table 12 (continued)

No	Paper	Study	Title	TC	Research Areas
19	Pözl-Stefanec and Geißler (2022)	Empirical	“Micro-Steps” on the Route to Successful Online Professional Development for Austrian Early Childhood Educators	6	Professional Development, Early Education
20	Javorcik et al. (2023)	Empirical	Microlearning in the Education of Future Teachers: Monitoring and Evaluating Students’ Activity in a Microlearning Course	6	Teacher Education, Monitoring
21	Isibika et al. (2021)	Empirical	Perceived Training Needs Assessment of Librarians in Tanzanian Academic Libraries Aimed at Introducing Microlearning Intervention to Training	5	Library Science, Training
22	Gorham et al. (2023)	Empirical	Analyzing Learner Profiles in a Microlearning App for Training Language Learning Peer Feedback Skills	5	Language Learning, Training
23	Romanenko et al. (2023)	Empirical	Microlearning as a New Method of Teaching Soft Skills to University Students	5	Soft Skills, Higher Education
24	Alias and Razak (2023)	Review	Exploring the Pedagogical Aspects of Microlearning in Educational Settings: A Systematic Literature Review	4	Educational Pedagogy, Microlearning
25	Naseem et al. (2022)	Empirical	The Outcomes of a Mobile Just-in-Time Learning Intervention for Teaching Bioethics in Pakistan	4	Bioethics, Mobile Learning
26	Mazlan et al. (2023)	Review	Exploring the Integration of Bite-Sized Learning: A Scoping Review of Research in Education and Related Disciplines	2	Bite-Sized Learning, Scoping Review
27	Lee et al. (2023)	Empirical	Development of a Bite-Sized Visual Interactive Study Aid for Carbohydrate Metabolism	2	STEM Education, Interactive Study Aid
28	Dolowitz et al. (2023)	Empirical	Iterative Design and Integration of a Microlearning Mobile App for Performance Improvement and Support for NATO Employees	2	Mobile Learning, Performance Improvement
29	Puah et al. (2021)	Empirical	Working Adults’ Intentions to Participate in Microlearning: Assessing for Measurement Invariance and Structural Invariance	2	Adult Learning, Measurement Invariance
30	Robles et al. (2023)	Empirical	Design of a Microlearning Framework and Mobile Application Using Design-Based Research	2	Mobile Application, Design-Based Research
31	Sedaghatkar et al. (2023)	Empirical	Enhancing Medical Students’ Knowledge and Performance in Otolaryngology Rotation through Combining Microlearning and Task-Based Learning Strategies	2	Medical Education, Task-Based Learning
32	Naser (2024)	Empirical	Tech-Enhanced Learning: Assessing the Impact of an Innovative Microlearning Module on Postgraduate Students’ Perceptions and Academic Progress	1	Higher Education, Tech-Enhanced Learning
33	Casebourne (2024)	Empirical	Left to Their Own Devices: An Exploration of Context in Seamless Work-Related Mobile Learning	1	Mobile Learning, Work-Related Learning

Table 12 (continued)

No	Paper	Study	Title	TC	Research Areas
34	Magri et al. (2022)	Empirical	Microlearning as a Tool for Scientific Dissemination: Thermal Analysis of the Portuguese Good Luck Rooster	1	Scientific Dissemination, Thermal Analysis
35	Borrego Ramírez and Ruiz Cansino (2023)	Empirical	Microlearning as an Alternative Teaching Model: Influence of COVID-19 in Latin America and the Caribbean	1	COVID-19, Teaching Models
36	Fialho et al. (2024)	Review	The Use of Microlearning in the Educational Field: An Overview of Worldwide Scientific Production	0	Educational Field, Scientific Production
37	Pham et al. (2024)	Review	Science Mapping the Knowledge Base on Microlearning: Using Scopus Database between 2002 and 2021	0	Science Mapping, Scopus Database
38	Rahbar et al. (2024)	Empirical	The Effectiveness of Social Media-Based Microlearning in Improving Knowledge, Self-Efficacy, and Self-Care Behaviors Among Adult Patients with Type 2 Diabetes: An Educational Intervention	0	Healthcare Education, Social Media
39	Rof et al. (2024)	Empirical	Exploring Learner Satisfaction and the Effectiveness of Microlearning in Higher Education	0	Higher Education, Learner Satisfaction
40	Corbeil et al. (2023)	Empirical	The Effect of Microlearning and Multimedia Design on Knowledge and Skills Acquisition of Students in E-Courses	0	Multimedia Design, Knowledge Acquisition
41	Auf (2024)	Empirical	Concurrent Vocational Education through Microlearning Approaches at Higher Education: Assessing the Capacity, Awareness, and the Will of Universities in Garowe, Somalia	0	Vocational Education, Higher Education
42	Lykova and Savelyeva (2024)	Empirical	The Place of Microlearning in Foreign Language Teaching Methodology: Historiographic Study	0	Foreign Language Teaching, Historiographic Study
43	Chamorro-Atalaya et al. (2024)	Review	Microlearning and Nanolearning in Higher Education: A Bibliometric Review to Identify Thematic Prevalence in the COVID-19 Pandemic and Post-Pandemic Context	0	Higher Education, Nanolearning
44	Al-Zahrani (2024)	Empirical	Enhancing Postgraduate Students' Learning Outcomes through Flipped Mobile-Based Microlearning	0	Mobile Learning, Flipped Classroom
45	Balasundaram et al. (2024)	Empirical	Microlearning and Learning Performance in Higher Education: A Post-Test Control Group Study	0	Higher Education, Learning Performance
46	Isibika et al. (2023)	Empirical	The Influence of User-Perceived Benefits on the Acceptance of Microlearning for Librarians' Training	0	Training, Library Science
47	Alshammari (2024)	Empirical	Video-Based Microlearning and the Impact on Programming Skills and Technology Acceptance	0	Programming Skills, Technology Acceptance
48	Aouad et al. (2024)	Empirical	Educating Primary Care Physicians About Eating Disorders: Pilot Data from a Microlearning Programme	0	Health Sciences, Disordered Eating
49	Ali et al. (2022)	Empirical	Intelligent System for Multi-Layer Lip-Reading-Based Microlearning	0	Higher Education, Technology Acceptance Model

Table 13 Summary of review studies

No	Author	Objective	Scope of Review	Methodology	Key Metrics	Main Findings	Limitations
1	Leong et al. (2021)	Review the trends of microlearning in publications and Internet searches	Global, covering various countries, education levels, and sectors	Bibliometric analysis, Google Trends analysis	Number of publications (2006–2019), Google Trends data on microlearning searches	Microlearning is a growing global topic in e-learning and mobile learning, with higher education being the most frequently mentioned education level. Language learning is a frequent subject. The research supply is aligned with the demand from internet users	Limited to Scopus and Google Trends data; does not account for other publication databases and search engines. Internet searches may not always reflect an academic or professional interest in microlearning
2	Fialho et al. (2024)	To understand the current state of scientific knowledge on the educational interfaces of microlearning	99 articles published worldwide since 2019, from 37 countries, primarily the United States	Exploratory, descriptive bibliometric search using the descriptors microlearning and education on Scopus; metadata processed using VOSviewer®	99 articles; significant growth in publications since 2019	Identified six thematic categories: formal educational innovations, microlearning in continuing education for health professionals, informal education, the potential of microlearning, digital mobile microlearning, and methodologies for implementation and evaluation	Limited production in the global south and eastern countries; there is a need for more research in those areas
3	Samala et al., (2023a, 2023b)	Explore the transformative potential of microlearning in the digital age and provide insights for future advancements	Focus on Scopus-indexed publications since 2015, showing a growing interest in microlearning research worldwide	The systematic review was aligned with PRISMA guidelines and combined with bibliometric analysis	Increase in publications related to microlearning since 2015	Microlearning is not only about delivering bite-sized content; it highlights the significance for Generation Z and digitally native generations accustomed to technology	The abstract does not specify limitations, possibly lacking a detailed exploration of specific challenges or gaps in microlearning research

Table 13 (continued)

No	Author	Objective	Scope of Review	Methodology	Key Metrics	Main Findings	Limitations
4	Pham et al. (2024)	Investigate the document types, volume, growth trajectory, geographic contribution, and coauthor relationships in the microlearning literature	297 Scopus-indexed documents from 2002 to 2021	Adaptation of PRISMA guidelines to assess document eligibility; descriptive statistics and science mapping conducted	297 documents were analyzed, and 178 conference papers (59.86%) were used	Increasing trend of microlearning publications; concentration of contributions from 15 countries (22.39%), yielding 68.8% of all documents; six key themes identified, including mobile learning design, MOOCs adaptation, and language teaching	The limited focus on educational levels was not explicitly analyzed, and potential biases in conference papers dominated the literature
5	Mazlan et al. (2023)	Investigate the potential of bite-sized learning in music education and its integration across various disciplines	Articles from Scopus and ERIC databases; focus on ICT, mathematics, and medicine, with emphasis on music education	Scoping review methodology to identify and analyze relevant literature	Articles identified from multiple databases focus on practical applications in music education	Bite-sized learning enhances accessibility and convenience, and reduces cognitive load; it can improve learning quality by making it enjoyable and tailored to learner interests. It aligns with 21st-century learning traits such as personalization	The review primarily focuses on music education; effectiveness in other subjects or contexts is not deeply explored
6	De Gagne et al. (2019b)	To conduct a systematic review of literature on microlearning in health professions education to identify key concepts and evaluate pedagogical outcomes	17 articles published between 2011 and 2018, covering various health disciplines (medicine, nursing, pharmacy, etc.)	Scoping review using databases such as PubMed, CINAHL, EMBASE, and others; assessment of pedagogical outcomes based on the Kirkpatrick model	3096 references were retrieved, and 17 articles were included after inclusion/exclusion criteria were applied	Microlearning positively affects knowledge and confidence in students of health professions, and various technologies are used. However, it also faces challenges such as pedagogical discomfort and technology inequalities	Lack of studies addressing higher-level outcomes; limited to English publications; potential gaps in assessing behavioral change and practice impact

Table 13 (continued)

No	Author	Objective	Scope of Review	Methodology	Key Metrics	Main Findings	Limitations
7	Sankaranarayanan et al. (2023)	To conduct a bibliometric analysis of microlearning literature to characterize and understand its themes and contexts	208 publications on microlearning were collected from Scopus	Quantitative topic modeling and qualitative content analysis methods	208 publications analyzed; themes identified	Four major themes were identified: design of microlearning, implementation as an instructional method, evaluation, and utilization of mobile devices. Findings emphasize the need for rigorous inquiry and expanding research contexts to include K-12 settings	Limited focus on specific educational levels (K-12) and mobile-based microlearning; more exploration is needed in other contexts
8	Alias and Razak (2023)	To conduct a systematic literature review to identify effective practices of microlearning in teaching and learning in higher education between 2014 and 2023	Higher education focuses on the pedagogical design of microlearning	PRISMA guidelines and systematic literature review are used to identify, evaluate, interpret, and analyze studies on microlearning practices	Pedagogical design, instructional flow	Two main themes: (1) Design of microlearning content and (2) Instructional flow. These provide insights for curriculum designers and educators on effective microlearning strategies	Limited to studies from 2014–2023, focus only on higher education, and do not explore corporate microlearning practices
9	Chamorro-Atalaya et al. (2024)	To explore the thematic trends of microlearning and nanolearning during and after the COVID-19 pandemic	Higher education, with a focus on applications in medicine, engineering, and skills development	Bibliometric review, quantitative approach	Prevalence of themes such as effectiveness, adaptation, and competency-building	The COVID-19 pandemic boosted microlearning adoption more than nanolearning, particularly in higher education. Key themes include adaptation and competency development in students. Identifies a gap in nanolearning adoption	Limited to the Scopus database, which may exclude other relevant studies; it focuses only on publications related to the COVID-19 pandemic, possibly overlooking trends unrelated to the pandemic

trends, practices, and key challenges in the implementation of microlearning across various educational and professional contexts while also providing guidance for readers seeking to explore this topic further.

Among the review studies of microlearning, De Gagne et al. (2019a) focused on the field of health professions education and conducted a comprehensive review of the application of microlearning. Leong et al. (2021) examined the development trend of microlearning from a holistic perspective. Sankaranarayanan et al. (2023) used bibliometric analysis to systematically inventory the application of microlearning in various fields. These literature reviews aim to summarize the development, application, challenges, and opportunities of microlearning. The findings outline the general picture of microlearning research and practice and identify research gaps and directions.

The content of the eight empirical studies is diverse and includes specific applications of microlearning in different educational contexts. For example, Park and Kim (2018) explored the design and development of microlearning in digital learning systems. Zarshenas et al. (2022) examined the impact of microlearning on nursing students' learning outcomes and self-efficacy. Fidan (2023) investigated the effectiveness of microlearning combined with flipped classrooms for teacher training, while McNeill and Fitch (2023) applied instructional design principles to analyze microlearning. Zheng (2021) explored the technical aspects related to microlearning, such as the classification of wireless network information. In addition to formal education, two empirical studies also addressed the application of microlearning in emerging fields (e.g., Conde-Caballero et al., 2024; Romero-Rodríguez et al., 2023).

These empirical studies examine the practical effects of microlearning in specific contexts by developing microlearning materials, conducting instructional experiments, and collecting qualitative or quantitative data. The value of these highly cited studies lies in the accumulation of empirical evidence on microlearning research, including the discovery of instructional designs and the effects of learning environments.

Discussion

Evolution of Keywords over Time

The analysis of keyword trends in microlearning research reveals key findings reflecting a shift in focus and the evolution of core themes from 2018 to August 2024. This analysis highlights significant changes in the understanding and application of microlearning. The key points identified in the keyword trends are shown in Fig. 8:

- Dominance of Core Concepts (2018–2019):** In the early period, dominant keywords such as “e-learning,” “bite-sized learning,” and “online learning” were used to define the concept of microlearning. This research emphasized how educational materials could be broken down into smaller, more digestible modules to enhance knowledge retention among learners (De Gagne et al., 2019a; Park & Kim, 2018).
- Shift to Digital Technology (2020–2022):** From 2020 to 2022, keywords such as “virtual learning,” “mobile applications,” and “adult learning” increasingly emerged, indicating a shift toward the use of digital technology in remote learning. The COVID-19 pandemic accelerated the adoption of mobile platforms and online learning, underscoring the critical role of technology in creating more flexible and accessible learning environments (Dolasinski & Reynolds, 2020; Lee et al., 2021; Susilana et al., 2022; Taylor & Hung, 2022).
- Increased Focus on Personalized Learning and Advanced Technology (2023–2024):** In 2023–2024, there was a notable increase in keywords such as “21st-century learning,” “self-efficacy,” and “active learning.” This trend signals a shift toward more active and personalized learning through the integration of technologies such as AI, machine learning, AR, and VR. These technologies create more immersive learning experiences that align with the needs of learners in the modern digital era (Bucaioni et al., 2024; Gill et al., 2024; Jauhiainen, 2024; Memarian & Doleck, 2024; Puah et al., 2021; Rapaka et al., 2025; Rejeb et al., 2024; Skalka et al., 2021; Zarshenas et al., 2022).
- Emerging Themes:** New keywords such as “learning analytics,” “social media,” and “mobile applications” indicate an increasing focus on innovative methods for

Fig. 8 Evolution of keywords in microlearning research (2018–August 2024)



tracking the effectiveness of microlearning through data analysis and the use of mobile platforms and social media to support learning (Chatti et al., 2012; De Witte & Chenier, 2023; Gonzalez & Chiappe, 2024).

The evolution of keywords illustrates how microlearning continues to adapt to meet the evolving demands of education and the corporate world. The dominance of terms such as “microlearning,” “bite-sized learning,” and “mobile learning” emphasizes the importance of delivering educational content in easily digestible, modular formats that are accessible in various locations. This highlights that micro-learning is increasingly recognized for its relevance across sectors in higher education and professional training, where flexibility and efficiency in information absorption play crucial roles. Furthermore, the increasing use of keywords such as “learning analytics” indicates that researchers and practitioners are increasingly interested in leveraging data to assess the effectiveness of microlearning interventions.

This keyword analysis provides a roadmap for understanding the core themes in microlearning and offers insights into future research directions in this field. As technology plays an increasingly significant role in shaping learning experiences, future research is likely to explore deeper integrations with AI (Samala et al., 2024a, 2024b, 2024c; Wu et al., 2024), mobile-first platforms (Criollo-C et al., 2024; Fitriyanti et al., 2023; Maulana, 2023; Samala et al., 2024a, 2024b, 2024c), and immersive technologies (Coban et al., 2022; Samala et al., 2024a, 2024b, 2024c; Scavarelli et al., 2021). Overall, the results of this keyword trend analysis indicate that research in the field of microlearning has evolved from an initial focus on the definition and validation of concepts to a broader adoption of digital technologies, ultimately exploring more sophisticated and personalized learning methods.

To further understand the development and depth of microlearning research, exploring the thematic evolution of core topics over time is essential. While keyword analysis offers insights into the most frequently researched concepts, thematic evolution delves deeper into how these concepts have developed, interacted, and transformed over the years. In the following section, we analyze how the main themes in microlearning evolved from 2018 to August 2024, highlighting the shift from foundational principles to more advanced and technology-based applications. This exploration provides an overview of the underlying patterns driving innovation and the ongoing expansion of microlearning as a learning method.

Thematic Evolution

From 2018 to 2024, microlearning research exhibited notable thematic evolution, mirroring changing priorities and

emerging needs across sectors like corporate training and higher education. Early studies primarily focused on defining and conceptualizing microlearning, emphasizing core ideas such as bite-sized learning and just-in-time learning. However, with the increasing integration of digital technologies, the research agenda progressively shifted toward exploring innovative applications and pedagogical enhancements.

Early Period (2018–2020): Focus on Concepts and Validation

In the early years, microlearning research concentrated on defining the concept and exploring its effectiveness in learning. Themes like “microlearning,” “chunked learning,” and “e-learning” were predominant. Many studies have aimed to understand how this modular learning approach could be applied across various contexts, particularly in higher education and technology-based training. The thematic map visualization indicated that these terms were often interrelated and dominated discussions in the academic literature.

Growth Period (2021–2022): Integration of Digital Technology

Entering the 2021–2022 period, microlearning themes began to evolve with stronger integration of digital technology. Keywords such as “mobile learning,” “gamification,” and “distance learning” started to emerge more frequently. This reflected a shift towards using more sophisticated technologies to support remote and data-driven learning, particularly in response to the COVID-19 pandemic. Corporate training also became more prominent in this research, alongside the growing need for efficient and flexible learning in increasingly decentralized work environments.

Technology and Personalization Period (2023–2024): Focus on AI and Adaptive Learning

In the 2023–2024 period, a significant change in the direction of microlearning research occurred. During this period, more advanced themes such as AI, machine learning, AR, VR, and adaptive learning began to dominate. This indicates a shift toward personalized learning with technology to create more immersive and individualized learning experiences.

Main Themes Supporting Flexibility and Scalability in Microlearning

The analysis results in Table 12 highlight various main themes and subthemes that support the flexibility, scalability, and practical application of microlearning across different educational and professional contexts. Prominent themes from this research include the application of microlearning in health education, higher education, and corporate

training, as well as the use of digital and mobile technologies as platforms to support modular learning.

Flexibility and Broad Applications across Sectors

1) Health Education.

Research by De Gagne et al. (2019a) and Zarshenas et al. (2022) emphasizes the importance of microlearning in health education, particularly for enhancing medical and nursing skills. In a dynamic sector like healthcare, where a rapid response to new knowledge is critical, microlearning offers a modular and flexible learning solution. This format allows timely training without disrupting daily work, supporting continuous learning in environments that require constant adaptation to new information.

2) Higher Education.

Conde-Caballero et al. (2024) and Romero-Rodríguez et al. (2023) explore the application of microlearning in higher education to enhance student engagement, particularly through social media platforms and collaborative learning. The flexibility of microlearning enables students to access materials anytime, anywhere, enriching their learning experience. Research by Matthews et al. (2014) and Polasek and Javorcik (2019) shows that microlearning can break down complex concepts into manageable segments, facilitating gradual understanding. Similarly, in language learning, Edge et al. (2011) and Khong and Kabilan (2022) demonstrated their effectiveness in reinforcing vocabulary and grammar through focused, short sessions.

3) Corporate and Professional Training.

Shabadurai et al. (2022) and Mujica-Luna et al. (2021) illustrate the use of microlearning in job-based training within the corporate sector. Its flexibility and scalability allow companies to provide brief, targeted training for employees, aligning skill development with the dynamic needs of the industry. In workplace learning, Dolasinski and Reynolds (2020), Emerson and Berge (2018), and Göschlberger and Bruck (2017) highlight how microlearning enables employees to learn without sacrificing productivity, as training materials can be accessed as needed.

Additionally, microlearning is widely applied in adult education and continuous learning, where Zaqoot et al. (2020) show that this format is handy for providing accessible training that fits busy schedules. In vocational education and professional development, Blau et al. (2020) and Zhang and West (2020) emphasize how microlearning efficiently facilitates the development of technical and practical skills without disrupting participants' work routines.

Microlearning in the Context of Digital and Mobile Technology

1) Mobile Learning.

Research by Samala et al., (2023a, 2023b) and Sedaghatkar et al. (2023) emphasizes the significance of mobile platforms in supporting microlearning. With easy access via mobile devices, learners can engage with microlearning content in various locations, allowing flexibility in learning time and place. However, technological infrastructure limitations in developing countries hinder wider adoption. In areas with unstable or limited internet access, the implementation of mobile learning may be less effective, particularly in rural regions, indicating that local contexts must be considered.

2) Multimedia and Interactive Applications.

Sung et al. (2023) highlight how multimedia and interactive elements in microlearning can enhance learner engagement. Attractive and interactive designs create more effective learning experiences in digital environments, helping students to understand content more deeply. This study has several limitations, particularly in learning contexts with limited access to multimedia technology, where the necessary hardware may only be available to some learners.

Scalability through Modular Design

1) Bite-Sized Learning.

The modular learning design explored by Mazlan et al. (2023) and Lee et al. (2023) emphasizes that presenting content in small portions allows learners to utilize brief study times efficiently. However, challenges arise when overly segmented learning materials result in a lack of comprehensive understanding of more complex topics. Specific skills or knowledge cannot be grasped solely through small chunks of information in higher education or professional training contexts. This limitation highlights that bite-sized learning is more than just a one-size-fits-all solution.

2) Flipped Classroom.

Studies by Fidan (2023) and Al-Zahrani (2024) illustrated how microlearning can support the flipped classroom model, which integrates synchronous and asynchronous learning. Learners can study materials independently before class, allowing classroom time to be utilized for in-depth and collaborative discussions. However, a drawback of this model is that students need access to materials before class, diminishing its effectiveness, particularly in developing countries with limited technological access. Therefore, while the flipped classroom supports professional development, adequate

technological infrastructure and digital education must be ensured.

Application of Microlearning for Professional Competency Development

1) Professional Development and Soft Skills.

Several studies, such as Romanenko et al. (2023) and Pölzl-Stefanec and Geißler (2022), highlight the importance of microlearning in developing professional competencies, including soft skills. The modular nature of microlearning offers flexibility for enhancing interpersonal and managerial skills crucial for career advancement. However, this approach may be less effective at cultivating more complex competencies, such as strategic leadership or decision-making. In these contexts, more structured and long-term learning approaches may be necessary, raising questions about the practical application of microlearning and its limitations in fostering greater soft skills.

2) Specialized Skills Training.

Research by Robles et al. (2023) and Dolowitz et al. (2023) emphasizes that microlearning is effective at enhancing specific technical skills, such as programming or task-based learning in professional settings. The flexibility provided allows training to occur in brief sessions, facilitating integration into work routines. However, in more complex industrial sectors, a structured curriculum may be required to ensure that all aspects of skills are comprehensively learned.

Factorial Analysis

Factorial analysis allows us to identify core themes and patterns in microlearning research by examining the relationships between various studies and their contributions to the field. As shown in Fig. 9, the result highlights how different research documents contribute to the themes that support flexibility and scalability in microlearning, particularly as applied in educational and corporate training environments.

The factorial analysis performed on the dataset revealed several significant clusters representing the main themes in microlearning research. The two main dimensions (dim1 and dim2) illustrate the distribution of documents across various thematic axes, helping to identify how each document relates to overall trends in microlearning.

The clustering result, as shown in Fig. 9, highlights several key documents that significantly contribute to the development of themes within microlearning. Each cluster represents groups of studies focused on specific aspects of microlearning, such as technology integration, content modularity, and adaptability to various learning environments. The dimensions (dim1 and dim2) indicate where each document resides within the thematic space, while TC reflects each document's impact within the academic community. The factor analysis results reveal that microlearning research focuses on flexibility and scalability, supported by technology and active learning methodologies, as illustrated in Fig. 10.

The result confirmed that the use of mobile and multimedia platforms enhances the scalability and flexibility of microlearning. The integration of these technologies into modular learning is becoming a prominent trend. In today's digital landscape, technology facilitates access to learning

Fig. 9 Keyword patterns

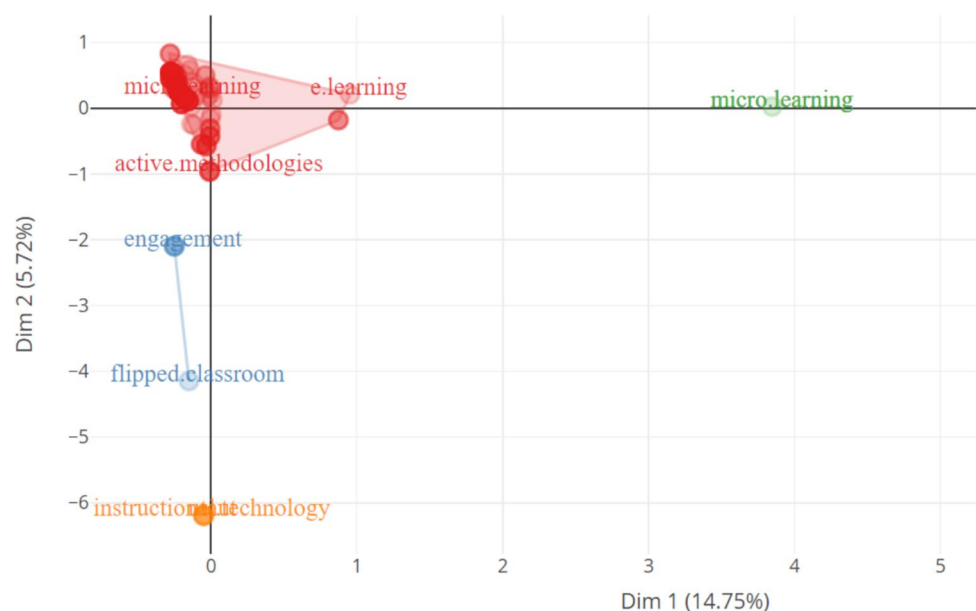
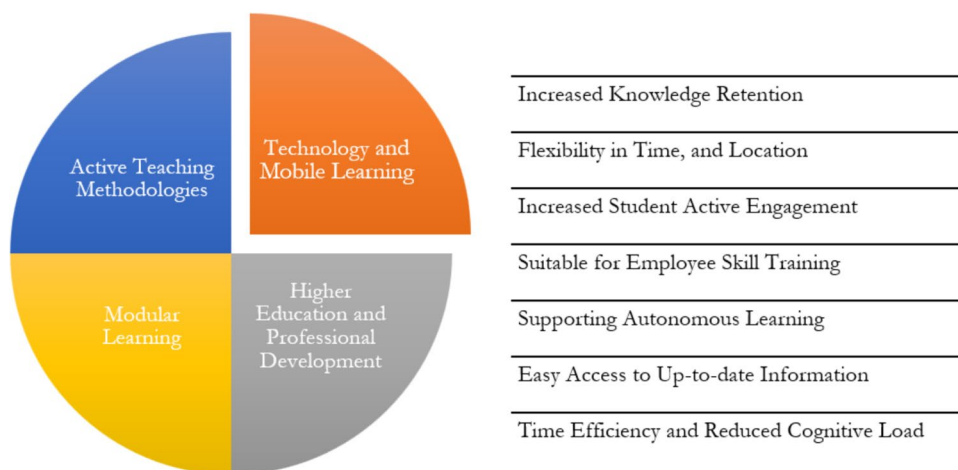


Fig. 10 Microlearning: active teaching methodologies, technology integration, modular learning, and professional development



anytime and anywhere, such as during waits or commutes. Digital platforms and mobile devices allow scalable learning that reaches a broader audience, providing essential flexibility in the modern era, particularly for professionals with busy schedules (Keengwe & Bhargava, 2014; Saikat et al., 2021).

Active teaching methodologies, such as flipped classrooms and gamification, play a central role in promoting student active engagement and motivation to learn (Leong et al., 2021; Sankaranarayanan et al., 2023). Our findings strengthen this assertion, as data from educational institutions indicate that these strategies significantly improve student retention. However, the effectiveness of these methods in the microlearning context depends on the material taught. For example, gamification proves more effective in technical skills development courses, while flipped classrooms have a greater impact on concept-based disciplines such as business or social sciences. This insight further validates the conclusions drawn from previous studies by highlighting that not all active learning strategies yield the same benefits across different learning environments.

Gamification, which incorporates game elements such as points, levels, and rewards, has enhanced student motivation and engagement. Microlearning addresses traditional challenges in maintaining student attention by transforming learning activities into more enjoyable experiences (Legaki et al., 2020; Murillo-Zamorano et al., 2021; Zainuddin et al., 2020). Additionally, the use of micro-credentials, which recognize skill achievements, can be amplified by gamification through badges that motivate learners to complete microlearning modules, creating a more interactive and personalized learning experience (Ahsan et al., 2023). This recognition makes students feel valued and motivated to complete microlearning modules, creating a more interactive and personalized learning experience. The integration of gamification and micro-credentials enhances student engagement and provides formal validation of the skills acquired, helping

students better prepare themselves to face challenges in the workforce. This combination fosters a sense of accomplishment and encourages lifelong learning, which is crucial for success in today's dynamic job market (Li & Ironsi, 2024).

Modular learning is central to microlearning, where content is broken into smaller, bite-sized units that are easier to digest. This approach simplifies the learning process and offers flexibility for learners to adjust their pace according to personal schedules (Dolasinski & Reynolds, 2020; Godsk & Møller, 2024; Kossen & Ooi, 2021). The modularity of microlearning also allows for greater personalization, enabling learners to choose the most relevant modules for their needs or interests, which is particularly beneficial in the workplace for skill enhancement.

Implementing microlearning in higher education and professional development has significantly impacted how individuals learn and advance in their careers. In higher education, microlearning simplifies the introduction of complex concepts through segmented, digestible content, enabling students to master topics gradually. For instance, short videos or interactive materials at universities facilitate more effective learning experiences (Barua & Lockee, 2024; Brankovic & Cantwell, 2022; Rincon-Flores et al., 2024).

In the workplace, microlearning has emerged as a powerful tool for employee training and development. Companies leverage microlearning modules for quick and effective training, particularly for new skills or adapting to new tools and technologies. This approach keeps companies competitive while enabling employees to enhance their skills in response to the dynamic industry demands (Beste, 2023; Hesse et al., 2019). Unlike previous studies that outlined general benefits, our analysis reveals specific industries and training programs adopting microlearning effectively, particularly in the technology and engineering sectors, for accelerated onboarding.

From this factor analysis, it can be concluded that microlearning offers a highly flexible and scalable solution in both formal education and professional training. The integration

of mobile technology and active teaching methodologies not only enhances engagement and interactivity but also tailors learning to individual needs. Modular learning addresses time constraints and learner capabilities, while its application in higher education and professional development paves new paths for acquiring relevant skills in the digital age. With emerging trends such as nanolearning, the future of microlearning appears promising, as it will lead to continual adaptation to diverse sectoral needs.

Challenges and Solutions

Microlearning has been globally recognized as an efficient, flexible, and scalable educational method, particularly in educational and professional training environments. However, its implementation in various countries, especially in the developing ones, faces significant challenges. Based on research conducted worldwide, the following are challenges in microlearning implementation and strategic solutions to overcome these barriers.

Technological Barriers

One of the greatest challenges in implementing microlearning in developing countries is access to technology (Díaz Redondo et al., 2021). Many developing nations still face limited technological infrastructure, including internet connectivity, device availability, and technology literacy. For example, internet access remains a significant issue in some rural or remote areas, hindering learners' ability to access typical online microlearning materials and utilize mobile devices (Saikat et al., 2021).

- a) Governments and international organizations need to collaborate to enhance Internet access and expand technological networks in rural and remote areas.
- b) Collaboration among governments, the private sector, and educational institutions is essential for providing affordable learning devices for learners in developing countries, such as smartphones or tablets.
- c) Developing offline-accessible microlearning content for learners in areas with limited internet access is crucial. This could include mobile applications that allow for content downloads for offline use. Solutions like MoodleBox serve as a prime option for addressing this challenge.

Pedagogical Limitations

Pedagogical challenges are a significant issue in the implementation of microlearning, particularly concerning effective learning design. Many educators in developing countries may lack experience in integrating microlearning into their

curricula or training programs. There is a need for a robust pedagogical framework to ensure that microlearning can be implemented effectively, enabling learners to achieve optimal learning outcomes (Alvarez, 2020; Boulos & Wheeler, 2007; Zawacki-Richter & Latchem, 2018).

- a) Educators are suggested to receive training on designing and implementing effective microlearning strategies, including the use of digital tools, interactive media, and microlearning-based teaching methods.
- b) Further research is needed to develop a pedagogical framework specifically designed for microlearning, particularly in the context of formal education and professional training in developing countries.
- c) The development of locally relevant microlearning content tailored to the needs of learners in developing countries ensures that it aligns with local social, cultural, and economic conditions.

Regional Disparities

The inequality between developed and developing countries in terms of access to learning resources and technology remains significant. Compared with developed countries, many developing nations face digital divides where access to both technology and education is limited compared to developed countries. This disparity can exacerbate global educational inequalities and hinder the widespread adoption of microlearning in developing regions (James, 2005; Mtebe & Raisamo, 2014; Van Dijk & Hacker, 2003).

- a) Encouraging collaboration between developed and developing countries to share resources, knowledge, and technology for microlearning implementation. Initiatives could include cross-country educational technology projects, teacher training programs, and exchanges of best practices.
- b) Governments in developing countries should adopt policies supporting the adoption of educational technology, including investment in digital infrastructure and training for educators.
- c) Supporting local educational institutions in adopting and adapting microlearning to meet local needs may include technical assistance, grant programs, or partnerships with international organizations to accelerate the adoption of learning technologies.

Cultural and Language Barriers

In addition to technological and pedagogical challenges, cultural and language barriers also significantly hinder the adoption of microlearning in developing countries. Microlearning content, which is often produced globally, may not

align with local cultural contexts, and the language used might not be understood by all learners in these regions (Eze et al., 2018; Henderson et al., 2017).

- a) Adapt microlearning content to fit local cultures and languages, helping learners better engage and understand the material.
- b) Microlearning content can be developed and translated into local languages to enhance accessibility and relevance.

Conclusion

Microlearning has emerged as a highly relevant educational approach in today's digital era. Its modular structure and brief, focused lessons effectively enhance knowledge retention, alleviate cognitive load, and increase learner motivation and engagement. These attributes make microlearning particularly well-suited for both formal education and professional training environments.

Key advantages of microlearning include flexible access via digital devices such as smartphones, tablets, and computers, facilitating self-directed learning that allows learners to customize their pace and focus according to individual needs. While especially pertinent to higher education, microlearning also proves valuable in workplace skill development. Its adoption by organizations and educational institutions highlights its practicality for delivering efficient training without compromising productivity.

Despite its advantages, several challenges hinder the widespread adoption of microlearning, particularly in developing countries. Key obstacles include limited access to technology, persistent digital divides, pedagogical constraints, and language or cultural barriers. Overcoming these issues requires coordinated efforts among governments, private sectors, and educational institutions to improve infrastructure, enhance educator training, and develop localized, culturally relevant content.

Looking ahead, the growth of microlearning is expected to accelerate through the integration of advanced technologies such as AI, AR, VR, and machine learning. These innovations promise to create more adaptive and personalized learning experiences. Furthermore, incorporating gamification and interactive elements is anticipated to boost learner engagement and enhance overall learning effectiveness.

Limitations and Future Work

This study has several limitations that need to be addressed. First, the bibliometric analysis was limited to data indexed in specific databases, such as Scopus and WoS, potentially

overlooking key publications from other sources. Additionally, while the study explores global trends in microlearning, research from developing countries remains underrepresented, which may result in an incomplete view of the unique challenges faced by these regions.

It is also important to note that the bibliometric analysis was based on a deliberately filtered corpus of 49 high-impact studies, as defined by the B + SLR framework and PRISMA inclusion criteria. Consequently, findings such as the number of influential authors, collaborations, and key thematic terms may appear relatively small. This is not a limitation of the study but a reflection of the emerging and consolidated nature of the high-impact microlearning research landscape. The “small numbers” observed indicate that a relatively small group of pioneering researchers, key institutions, and core themes currently dominate the field, providing insight into the intellectual center of gravity rather than the full breadth of all publications.

Future research should focus on creating practical guidelines for educational institutions and companies while considering local factors such as culture, language, and infrastructure. As technologies such as AI, AR, and VR increasingly influence digital learning, upcoming studies could explore how these innovations can enhance personalized, interactive, and immersive microlearning experiences to boost effectiveness.

Furthermore, as online microlearning expands, future work is needed to investigate ways to improve social interaction and collaboration within this learning model. Integrating collaborative learning elements can enrich learning experiences. Addressing these limitations and exploring new opportunities will deepen the understanding of microlearning potential and broaden its application in various educational and professional contexts.

Author's Contribution **Agariadne Dwinggo Samala:** Conceptualization, Methodology, Investigation, Software, Formal Analysis, Writing – Original Draft, Visualization, Supervision. **Aidi Ahmi:** Methodology, Formal Analysis, Validation, Writing – Review & Editing, Supervision. **Febri Prasetya:** Conceptualization, Methodology, Investigation, Writing – Review & Editing. **Georgios Lampropoulos:** Validation, Writing – Review & Editing. **Soha Rawas:** Validation, Writing – Review & Editing. **Fadhli Ranuharja:** Conceptualization, Investigation, Writing – Review & Editing. **Kai-Yu Tang:** Validation, Writing – Review & Editing. **Mustafa Fidan:** Methodology, Formal Analysis, Writing – Review & Editing. **Putra Jaya:** Writing – Review & Editing. All authors have thoroughly reviewed, critically evaluated, and approved the final manuscript.

Funding The authors would like to thank Lembaga Penelitian dan Pengabdian Masyarakat Universitas Negeri Padang for funding this work with a contract number: 2048/UN35.15/LT/2025.

Data Availability The data used in this research are available upon request. Interested parties can access the dataset by contacting the corresponding author. The dataset includes all relevant documents

and information analyzed during the study, ensuring transparency and reproducibility of the research findings. The dataset can be accessed at the following link: https://osf.io/j4k92/?view_only=89ba572bbb4c4376af1cf1438222f62d

Declarations

Ethical Approval Ethical approval was not needed for this study because it involved the analysis of publicly available bibliometric data and did not include any human or animal subjects.

Informed Consent Not applicable.

Statement Regarding Research Involving Human Participants and/or Animals This research did not involve any human participants or animals.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process During the preparation of this work, the author(s) used OpenAI's ChatGPT to improve the readability and language of the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

Competing Interests The authors declare that they have no competing interests.

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